
KP35403 NETWORK MANAGEMENT AND MONITORING**SEMESTER 2, SESSION 2024/2025****SECTION 1****GROUP ASSIGNMENT: PING PALS****TITLE:**

Final Report : Design and Evaluation of a Network Monitoring Solution for a Small to Medium Enterprise (SME) Using PRTG Network Monitoring.

PREPARED FOR:

TS. NORDIN SAAD

No.	NAME	MATRIC NUMBER
1.	MUHAMMAD DANIEL BIN ROSLEY	BI22110117
2.	NORMA IRWANY BINTI GHAZALI	BI22110166
3.	ALDRIN ANAK PETER	BI22110473
4.	AZHAR BIN ABDUL KADIR	BI22110202

TABLE OF CONTENTS

1. Introduction and Literature Review.....	3
1.1. Introduction.....	3
1.2. Importance of Network Monitoring for SMEs.....	3
1.3. Overview of Existing Tools and Related Studies (Literature Review).....	5
1.3.1. PRTG Network Monitor for Real-Time Monitoring and Network Security.....	5
1.3.2. Comparative Analysis of PRTG and Wireshark for Network Monitoring.....	5
1.4. Role of PRTG Monitor in Network Management.....	7
2. System Design: Architecture, Protocol and MIBs.....	8
2.1. Network Architecture Overview (Logical Topology).....	7
2.2. Network Architecture Overview (IP Address Table).....	8
2.3. Protocols Used in Monitoring.....	11
3. Implementation: Tool Setup and Configuration.....	14
3.1. Tool Used For Simulation and Monitoring.....	14
3.2. Integration with VirtualBox and GNS3.....	15
4. Evaluation and Results.....	33
4.1. PRTG Monitoring Scenarios.....	33
4.1.1. DHCP Service Monitoring.....	33
4.1.2. DNS Service Monitoring.....	35
4.1.3. ICMP Service Monitoring.....	37
4.1.4. UDP Service Monitoring.....	38
4.1.5. Packet Sniffer Service Monitoring.....	40
4.2. Analysis of Monitoring Data.....	41
4.3. Observations and Performance.....	43
5. Conclusion and Future Work.....	45
5.1. Summary of Findings.....	45
5.2. Recommendations for Future Improvements.....	45
6. References.....	46

1. Introduction and Literature Review

1.1. Introduction

In an era dominated by digital communication and data exchange, the reliability and efficiency of network infrastructure have become paramount for businesses of all sizes. For Small to Medium Enterprises (SMEs) that lean heavily on IT stacks need a steady, safe and quick network to keep doors open and stay ahead of rivals. However, a lack of funding and technical resources makes it challenging for many SMEs to manage and monitor their networks effectively. As a result, their activities may therefore be disproportionately affected by network-related problems including latency, outages and security breaches.

With PRTG Network Monitor being the key tool, this research presents the design and analysis of a network monitoring solution specifically tailored to the requirements of SMEs. PRTG is a widely used, user-friendly network monitoring software developed by Paessler AG, capable of tracking network performance, uptime, bandwidth usage and other critical metrics in real time through customizable sensors and dashboards. This project utilizes PRTG to demonstrate how SMEs can develop performance management, enhance security awareness and gain a deeper understanding of their network activities without requiring substantial investment or advanced technical expertise. The study also explores the theoretical underpinnings of network monitoring, evaluates current methods and highlights the role of PRTG in addressing the unique challenges faced by SME environments.

1.2. Importance of Network Monitoring for SMEs

Network monitoring is essential for Small to Medium Enterprise (SME) because it enables early detection of network issues, enhanced security management and assured maximum system performance. SMEs, with limited IT capabilities, are very prone to business interruption by slow links, outages, or intrusions in the network. With ongoing monitoring of network usage, SMEs are able to detect and correct issues upfront before they interfere with business processes. This avoids costly downtime, saves confidential data and provides for seamless workflow. Ultimately, effective network monitoring

boosts productivity, protects the business's image and improves overall operational efficiency.

One of the primary advantages of network monitoring is its ability to detect performance issues, such as slow connection speeds or intermittent connectivity before they escalate into significant disruptions. By catching such problems at an early phase, SMEs can initiate remedial measures that can prevent costly downtime, which could undermine productivity, revenues and customer satisfaction otherwise. For instance, when a server is about to fail, monitoring software can alert IT staff so they can act in time and prevent the entire system from going down.

In addition to performance management, network monitoring enhances the security posture of SMEs by enabling real-time detection of threats such as malware infections, unauthorized access, or suspicious traffic patterns. Early identification of such anomalies allows organizations to respond swiftly and effectively, reducing the risk of data breaches, financial loss, or reputational damage. For instance, unusual traffic behavior flagged by a monitoring tool could indicate a potential cyberattack, prompting immediate countermeasures to isolate and neutralize the threat.

Network optimization is yet another key advantage. SMEs are able to detect bottlenecks, remove inefficiencies and streamline resource usage by keeping track of bandwidth usage and traffic trends in real time. The result is faster data transfer speeds, quicker applications and an improved overall user experience. For instance, monitoring provides early warning when a specific application is using too much bandwidth, enabling the appropriate changes to be implemented to restore network balance.

In addition, by reducing the number of unplanned outages and the need for emergency IT assistance, network monitoring lowers spending. It also makes capacity planning and adding infrastructure easier by offering insights into the use of resources, making it easier to plan. For instance, the company can consolidate operations or delay the acquisition of unneeded hardware if a server is not heavily used.

Lastly, a properly maintained and monitored network guarantees that employees are always able to access the digital tools and systems they require. This makes day to day work run smoothly and boosts employee satisfaction and productivity. For example,

network monitoring can detect and fix the root causes of sluggish performance in a customer relationship management (CRM) system that a sales team depends on, allowing the team to work efficiently.

In conclusion, by the strategic investment in network monitoring, SMEs can enhance the reliability, security, functionality and performance of their IT infrastructure. By enabling proactive issue detection and resolution, it can assist these enterprises to protect sensitive information, maintain operational effectiveness and compete more effectively in the highly competitive digital era.

1.3. Overview of Existing Tools and Related Studies (Literature Review)

1.3.1. PRTG Network Monitor for Real-Time Monitoring and Network Security.

Recent research literature highlights the increasing significance of integrated and user-centered network monitoring systems in enhancing cybersecurity and operational efficiency, especially in demanding settings such as schools. In university campus network management, Fathima and Devi (2024) suggested a real-time monitoring and threat detection system by combining the Paessler PRTG Network Monitor and Sophos Firewall. The combination provided real-time visual recognition of network traffic, quick identification of abnormal patterns and the generation of automated alerts with dynamic thresholds. The system also merged performance data with anomaly detection capabilities around issues such as latency, traffic bursts, device failure, and potential cyber-attacks via the SNMP, NetFlow, and sFlow protocols.

The research validated PRTG sensor-based all-encompassing monitoring platform apt for large-scale environments, hence highly relevant to Small to Medium Enterprises (SMEs) requiring cost-effective and scalable solutions. The research documented PRTG's simple-to-use dashboards, alert customizability and graphical tools enabling administrators to identify performance errors and security threats in a timely fashion. Apart from this, integration with Sophos Firewall enabled advanced filtering, user-based access control and granular traffic classification to mitigate threats such as Distributed Denial of Service (DDoS) and attempted unauthorized accesses.

Applying this framework on a university network, the study proved that PRTG is not just a performance monitoring tool but also a powerful solution to enhancing the network's security. The findings validate the application of PRTG in SME infrastructures where budgets for IT are normally constrained but a genuine requirement for trusted, real-time network observation and safeguarding exists.

1.3.2. Comparative Analysis of PRTG and Wireshark for Network Monitoring.

The understanding of what SMEs would realistically expect from network monitoring solutions requires appreciating the significant capabilities and limitations of some of the most common tools. Good insights from a recent comparative study by Alizan and Zolkipli (2024) offer useful information on the practical use of two specific network monitoring and analysis tools, Paessler Router Traffic Grapher (PRTG) and Wireshark. In light of their decisions relating to the network monitoring and analysis requirements of the company's business, the practical outcomes of the case study are valuable for anyone reviewing the use of Wireshark and PRTG in SMEs.

Wireshark is a versatile and open-source packet analyzer that aids deep cycle network diagnostics and forensic analysis. Wireshark excels in packet-level inspection, protocol decoding and network troubleshooting, but requires advanced knowledge and or skills for event security analysts who use the tool. Wireshark also does not have any built-in alerting, or any automated mapping function, or dashboards that can be tailored to the user's preference or to the network monitoring environment, which limits its potential practical application. This is particularly true when we talk about accessing some of the use cases of Wireshark for more proactive network monitoring of larger opposing or vast networks, that an operator manually observes as it is impossible to know about all aspects at any one time, even when you use Wireshark. The other problem with the normal use of Wireshark in an SME is that it has the steepest learning curve of all analysis tools and it requires the user to use it casually, which means capturing local traffic to ensure the practical uses of Wireshark.

PRTG is a much more robust and scalable network monitoring solution. In comparison to being able to use Wireshark, PRTG is better suited to SMEs since it is easy for the user

to establish. It has real-time monitoring and alerting capabilities. PRTG is widely used by network administrators, especially local SMEs. The comparative report provides good use cases with PRTGs mapping visualizations, bandwidth utilization measurement and custom dashboards. The benefits of PRTG are that the administrator can quickly see performance problems, the differences in status for monitoring system health and receive notifications without any special training. PRTG is simple to deploy, centralized and will provide administrators with a look at the network devices as a whole, which is helpful for companies using limited IT resources.

To conclude, as stated by the authors, Wireshark is still a unique deep packet analysis tool, but PRTG better meets the operational requirements of companies utilizing an accessible and real-time network monitoring solution. The authors have suggested an option of utilizing both as applicable, where PRTG could be used for monitoring the infrastructure broadly, while Wireshark is used as a deeper tool to investigate issues.

1.4. Role of PRTG Monitor in Network Management

PRTG Network Monitor is a handy application delivering efficient and simple network monitoring, especially for Small and Medium-Sized Businesses (SMBs) that tend to have hardly any technical personnel. PRTG is a Paessler AG uptime-monitoring solution packed with features that allows companies to maintain networks in top working order, identify faults before they occur, and enhance performance from a single system.

Real-time monitoring is one of the most important features provided by PRTG. It enables administrators to recognize issues such as service unavailable time, increasing latency, or hardware failure before they become critical. The tool uses dynamic dashboards and flexible sensors to send real-time alerts so that technical staff have ample time to solve issues and minimize downtime.

PRTG also provides an overview of all the networked devices. Via supported protocols in the form of SNMP, WMI, NetFlow, sFlow, and packet sniffing, the software can monitor switches, routers, servers, virtual machines, and applications simultaneously. The overview is easy to use when running the network and reduces the number of specialized solutions that are necessary, a feature which is especially useful for SMEs with smaller IT personnel.

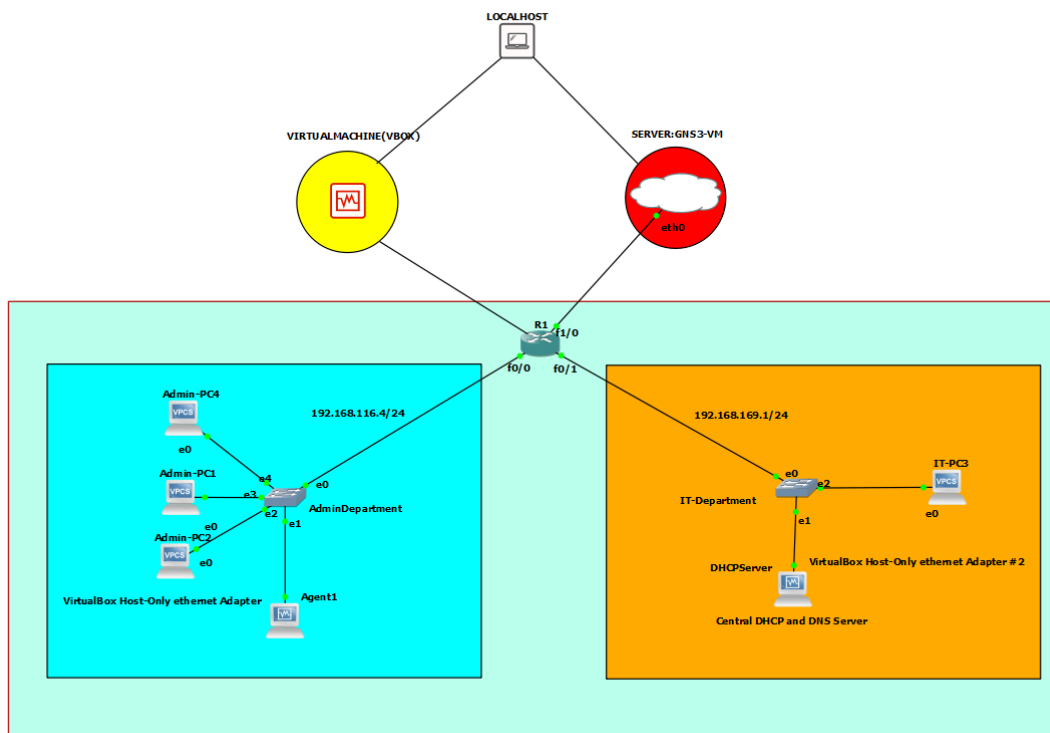
Trend analysis and reporting is yet another vital task carried out by PRTG. Such software keeps the history of the performance level of past time periods, and these figures are utilized for calculating capacity planning, ongoing problems, and to ascertain the impact of previous network downtimes. This assists in decision-making for future upgrades or configurations.

Secondly, PRTG also serves as a significant factor in improving network security. It identifies unusual traffic, unauthorized access, and potential attacks by examining network behavior over time. With adjustable thresholds and notification parameters, administrators can respond to security threats in advance and maintain data integrity.

PRTG is a cost-effective and versatile solution for SMEs to provide a sound, secure, and efficient network infrastructure. PRTG, with its easy-to-use functionality and mature monitoring features, is the ideal solution for those who require serious network management without enterprise complexity and expense.

2. System Design: Architecture, Protocol and MIBs

2.1. Network Architecture Overview (Logical Topology)



Network Simulation (Small Office Network)

2.1.1. Show an image of Logical Topology for Simulation of Small Network Office

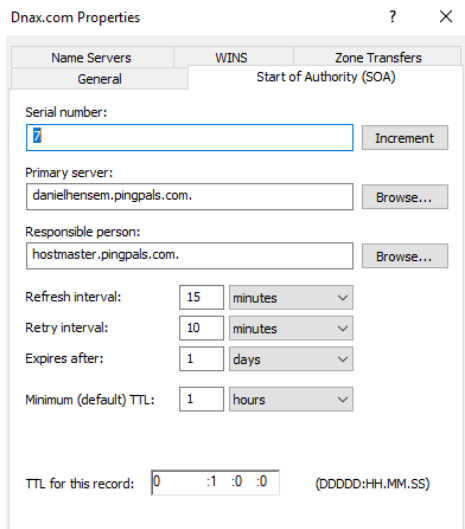
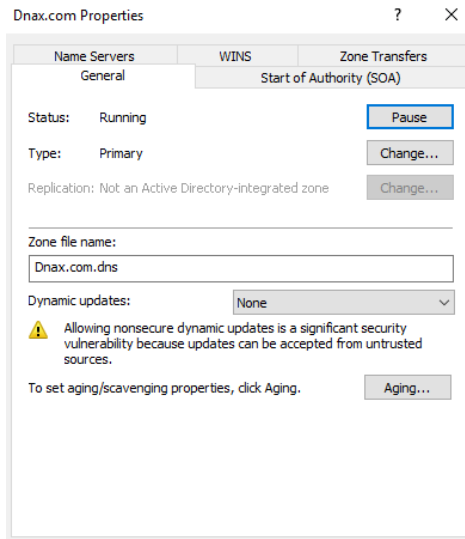
2.2. Network Architecture Overview (IP Address Table)

Device Name	Interface	IP Address	Subnet Mask	Network	Description
Router R1	FastEthernet 0/0	192.168.116.4	255.255.255.0	192.168.116.0/24	Gateway for Admin network
Router R1	FastEthernet0/1	192.168.169.1	255.255.255.0	192.168.169.0/24	Gateway for IT/DHCP network
DHCP Server	VirtualBox Host-Only ethernet Adapter #2	192.168.169.22	255.255.255.0	192.168.169.0/24	Windows Server 2019 (DHCP/DNS)
Agent1 (VM)	VirtualBox Host-Only ethernet Adapter	DHCP (192.168.116.101)	255.255.255.0	192.168.116.0/24	Monitoring/Testing node
GNS3 VM (eth0)	eth0	192.168.154.3	255.255.255.0	192.168.154.0/24	Internal web/file server
IT-PC3 (VPCS)	eth0	DHCP (192.168.169.103)	255.255.255.0	192.168.169.0/24	IT department client
Admin-PC1 (VPCS)	eth0h	DHCP (192.168.116.100)	255.255.255.0	192.168.116.0/24	Admin department client
Admin-PC2 (VPCS)	eth0	DHCP (192.168.116.113)	255.255.255.0	192.168.116.0/24	Admin department client
Admin-PC4 (VPCS)	eth0	DHCP (192.168.116.102)	255.255.255.0	192.168.116.0/24	Admin department client

DHCP Address Pool

Scope Name	DHCP Network Address	IP Address Start	IP Address End
SCOPE-13	192.168.116.0/24	192.168.116.100	192.168.116.254
SCOPE-12	192.168.169.0/24	192.168.169.100	192.168.169.254

Domain Name Server Pool(DNS)

Name	Type	Data
Dnax.com	Start of Authority(SOA)	
Dnax.com	Name Server(NS)	

2.3. Protocols Used in Monitoring

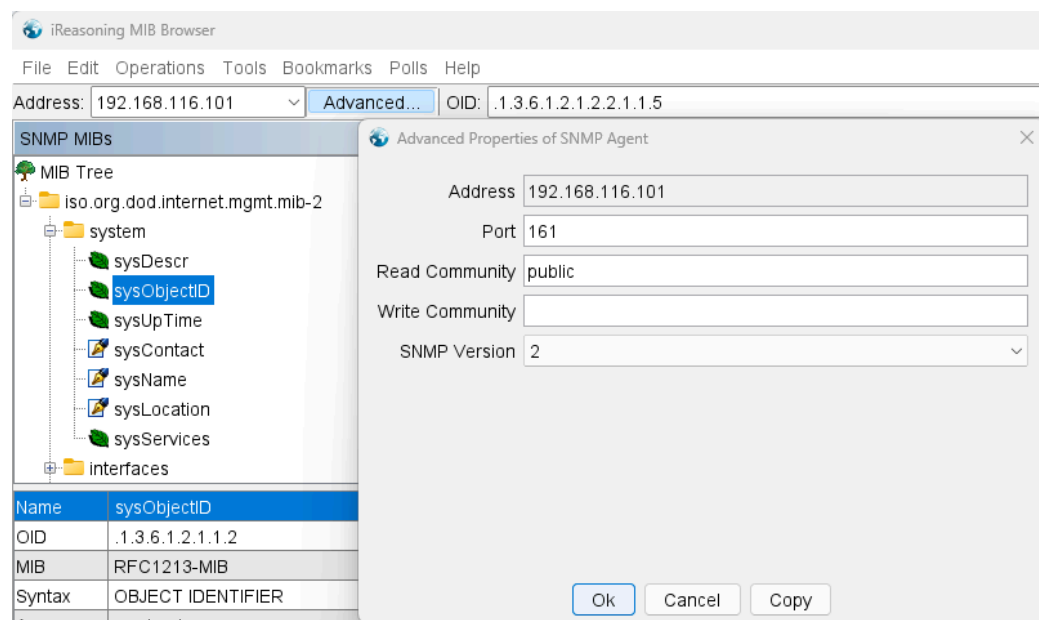
Protocol Used	Description
DHCP (Dynamic Host Configuration Protocol)	Tests a DHCP server can assign an IP address. Use it to simulate a DHCP request and confirm the server responds.
DNS (Domain Name Server)	Domain name using the target DNS server (pingpals.com). Verifies DNS service is up.
ICMP (Internet Control Message Protocol)	Pings the target device and reports latency and packet loss. Basic availability check.
UDP (User Datagram Protocol)	connectionless transport layer protocol used in computer networks. It allows data to be sent without establishing a connection between the sender and receiver.
Packet Sniffer (uses WinPcap or Npcap driver)	Captures all traffic visible to the NIC (interface) it's monitoring

2.4. Management Information Base (MIB) Concepts and Relevance

Management Information Base (MIB) is a hierarchical database used by SNMP-enabled devices to store and organize information about the device's operational status and configuration. Each data point in the MIB is identified by an Object Identifier (OID), which allows SNMP managers to query specific metrics.

2.4.1. How MIB Works with SNMP

MIB files act as dictionaries for SNMP communication, allowing the SNMP manager (like PRTG) to understand what kind of data it can request from a device. Without MIBs, SNMP would not be able to interpret the OIDs correctly.



2.4.1.1. Show an image of SNMP OID Values Retrieved from MIB

It shows several values retrieved from the device (Agent1), including system uptime, contact information, and interface indexes. These values are extracted based on specific Object Identifiers (OIDs) defined in the MIB file. The SNMP manager uses this structure to decode the data accurately, confirming that the SNMP service is active and the device is responding correctly to queries.

The screenshot shows the iReasoning MIB Browser interface. The address bar is set to 192.168.116.101 and the OID is .1.3.6.1.2.1.2.1.1.5. The left pane shows the MIB tree with 'sysObjectID' selected. The right pane displays a 'Result Table' with the following data:

Name/OID	Value	Type	IP:Port
sysContact.0		OctetString	192.168.116.10...
sysLocation.0		OctetString	192.168.116.10...
sysObjectID.0	.1.3.6.1.4.1.311.1.1.3.1.1	OID	192.168.116.10...
ifIndex.1	1	Integer	192.168.116.10...
ifNumber.0	15	Integer	192.168.116.10...
sysUpTime.0	17 minutes 40.87 seconds (106087)	TimeTicks	192.168.116.10...
ifIndex.2	2	Integer	192.168.116.10...
ifIndex.3	3	Integer	192.168.116.10...
ifIndex.4	4	Integer	192.168.116.10...
ifIndex.5	5	Integer	192.168.116.10...
sysServices.0	76	Integer	192.168.116.10...
sysName.0	17 minutes 40.87 seconds (106087)	OctetString	192.168.116.10...

Below the table, a detailed view of 'sysObjectID' is shown:

Name	sysObjectID
OID	.1.3.6.1.2.1.1.2
MIB	RFC1213-MIB
Syntax	OBJECT IDENTIFIER
Access	read-only
Status	mandatory

2.4.1.2. Show an image of SNMP Query Results Displayed via MIB

Image shows the successful retrieval of SNMP data from a device using the iReasoning MIB Browser. Several OIDs are displayed in the result table, including sysUpTime.0, sysName.0, and sysServices.0, each with corresponding values. This shows SNMP agent on the monitored device is actively responding to requests and providing detailed system metrics as defined in the MIB. The retrieved data verifies operational status such as uptime duration and service availability, proving the effectiveness of MIB in translating low-level SNMP data into human-readable information.

Add Sensor to Device Probe Device [127.0.0.1]

The screenshot shows the 'Add Sensor to Device' configuration window. The 'Basic Sensor Settings' section includes 'Parent Tags' (empty), 'Tags' (snmplibrarysensor), and 'Priority' (★★★★☆). The 'SNMP Library Specific' section shows the 'Library' path as 'C:\Program Files (x86)\PRTG Network Monitor\snmplibs\SNMPv2.oidlib'. Below this is a table of available sensors:

☐	MIB Module	Category	Name
☒	SNMPv2-MIB	snmp	snmp out traps
☒	SNMPv2-MIB	snmp	snmp out get responses
☐	SNMPv2-MIB	snmp	snmp out set requests
☐	SNMPv2-MIB	snmp	snmp out get nexts

2.4.1.3. Show an image of SNMP Custom Sensor Configuration in PRTG



2.4.1.4. Show an image of SNMP Sensor Monitoring Output

Image above displays the monitoring output of an SNMP sensor configured using the SNMPv2-MIB module in PRTG. This interface presents a real-time graph of SNMP metrics such as SNMP out traps and SNMP out get responses, providing visual insight into the device's SNMP activity over time.

3. Implementation: Tool Setup and Configuration

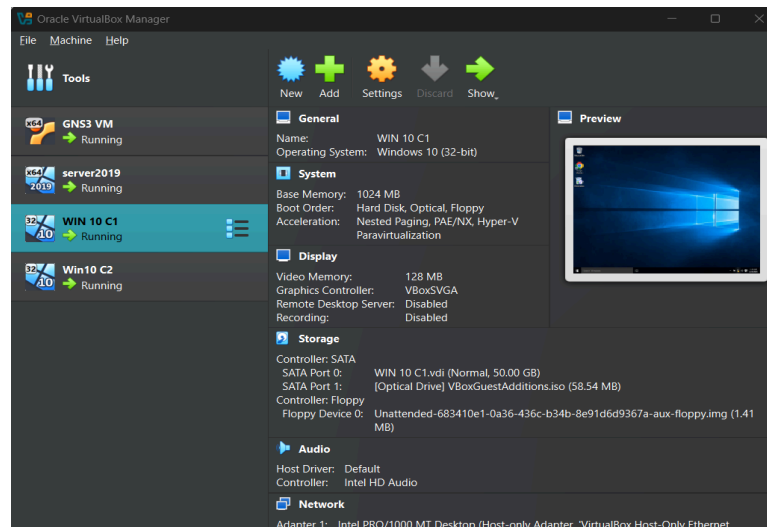
3.1. Tool Used For Simulation and Monitoring

Tool Used	Description
GNS3	Network simulation tool used to design, configure, and test virtual network topologies, including routers, switches, VPCS, and cloud connections.
Virtual Box	Virtualization platform used to run Windows Server 2019, Agent1, and other virtual machines that act as DHCP, DNS, or monitoring clients within the network.
PRTG	Network monitoring tool used to observe system health, traffic flow, availability, and performance of devices through sensors for protocols like ICMP, DNS, DHCP, and HTTP.
Laptop	Main host machine used to run GNS3, VirtualBox, and PRTG, acting as the central platform for configuring, simulating, and monitoring the

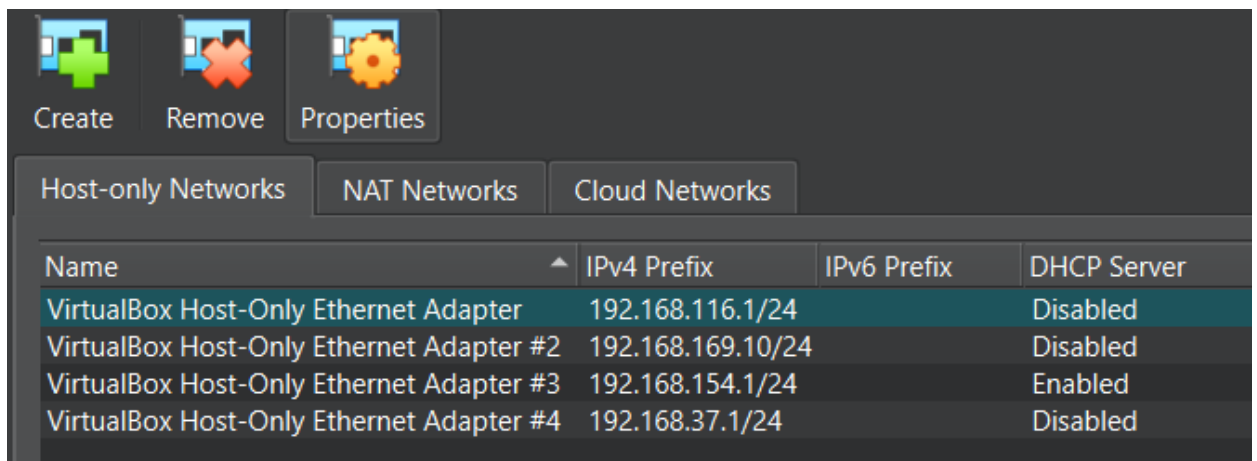
	virtual network.
Paessler MIB Importer V3	Import and convert MIB files into .oidlib format, enabling custom SNMP sensors to be added in PRTG for advanced monitoring.
MIB Browser	Query SNMP devices, browse MIB objects, and verify SNMP communication by retrieving real-time values from network agents.

3.2. Integration with VirtualBox and GNS3

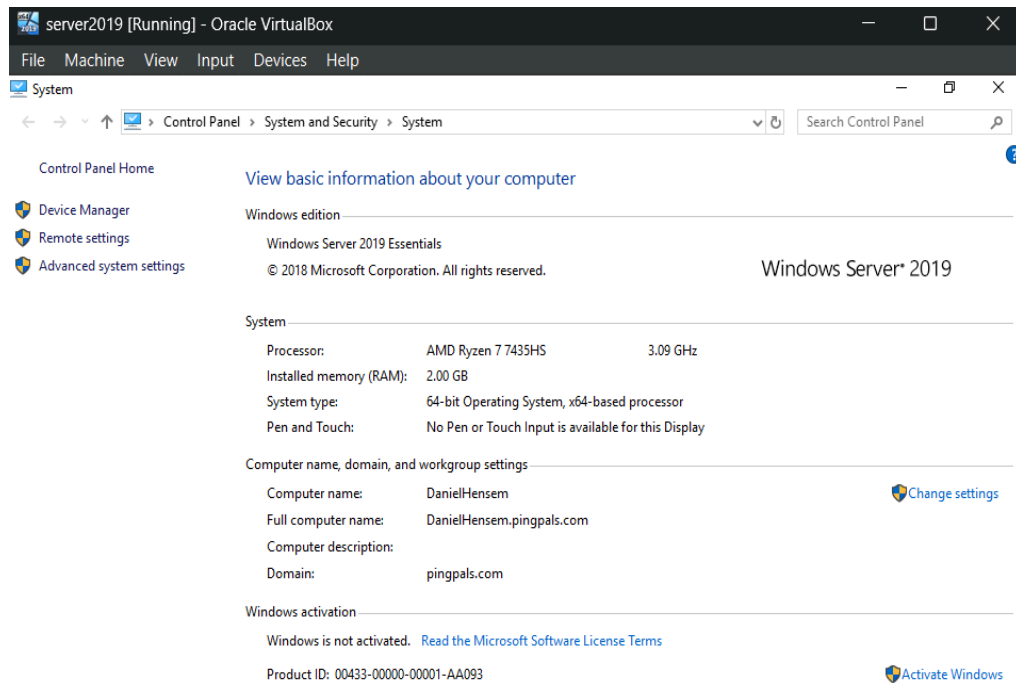
VirtualBox and GNS3 to create a hybrid simulation environment. Virtual machines running Windows OS were connected to GNS3 using host-only adapters and cloud nodes, enabling communication with simulated network devices. This setup allowed the testing of services such as DHCP, DNS, and network monitoring through PRTG in a realistic and controlled environment.



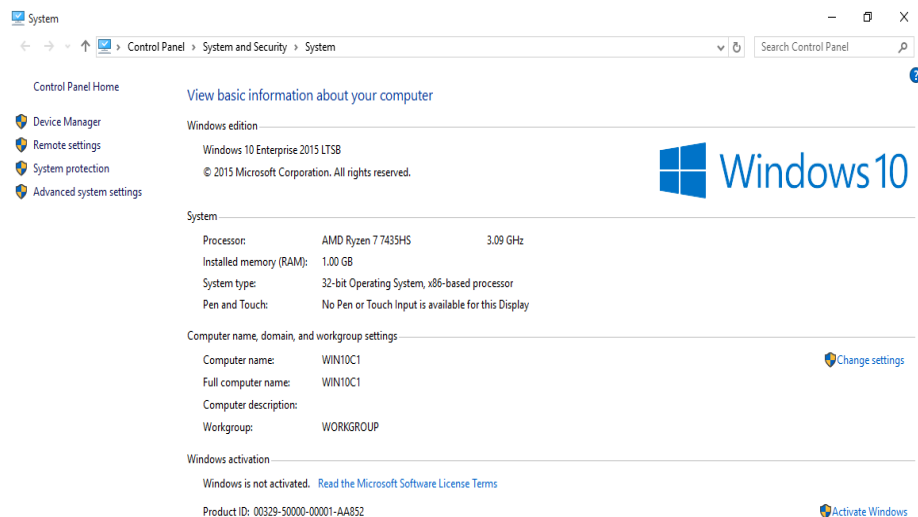
3.2.1. Show an image of Window Server 2019 and Client PC (Window 10 Agent for monitoring)



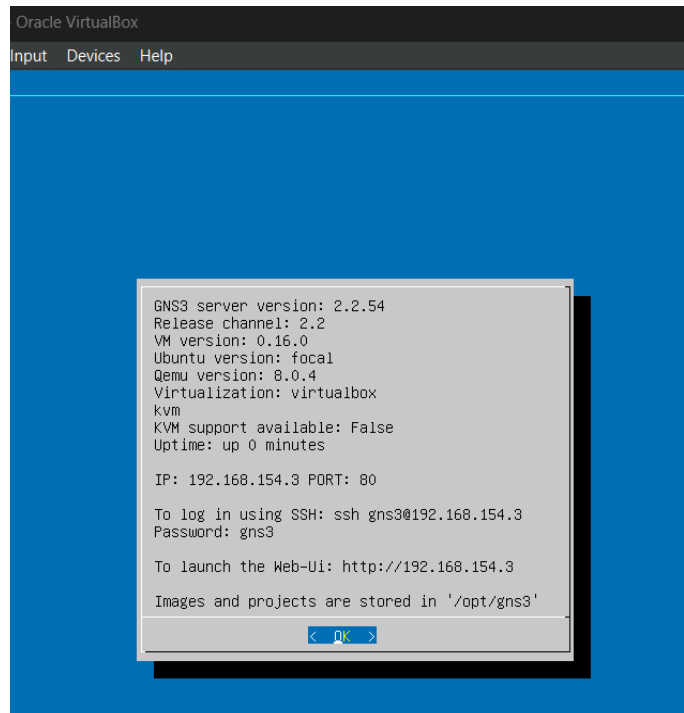
3.2.2. Show an image of VirtualBox Host-Only Network settings, which displays the configured host-only adapters used to connect virtual machines to the GNS3 network



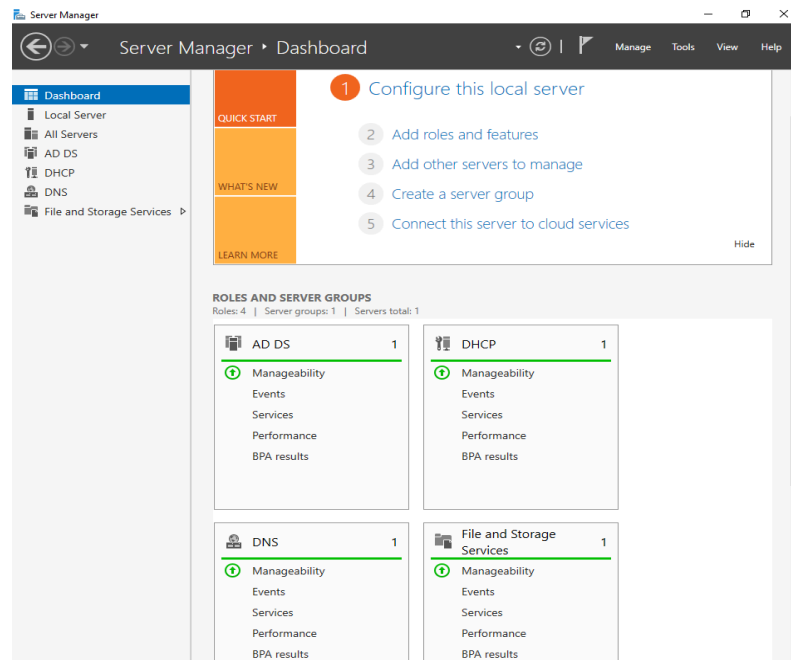
3.2.3. Show an image of Windows Server 2019 system properties, which displays key details such as Window Server 2019 Essentials and domain configuration. This server acts as the central DHCP and DNS server in the simulation. It is configured under the domain pingpals.com, aligning with the role of the server in managing network services within the GNS3 and VirtualBox-integrated environment.



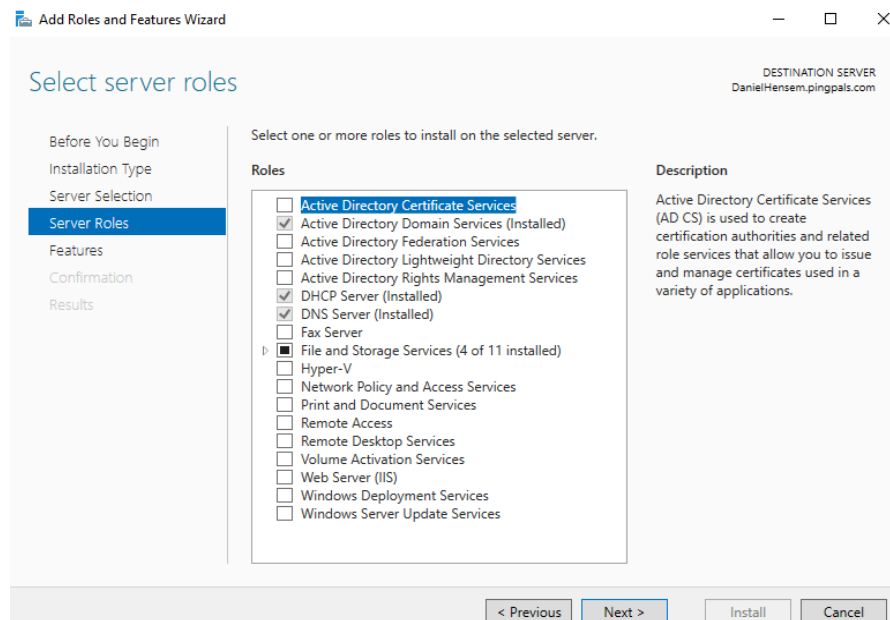
3.2.4. Show an image of Windows 10 that acts as client PCs to operate DHCP protocol and DNS server. This client device obtains its IP configuration dynamically from the DHCP server and uses the DNS server for domain name resolution.



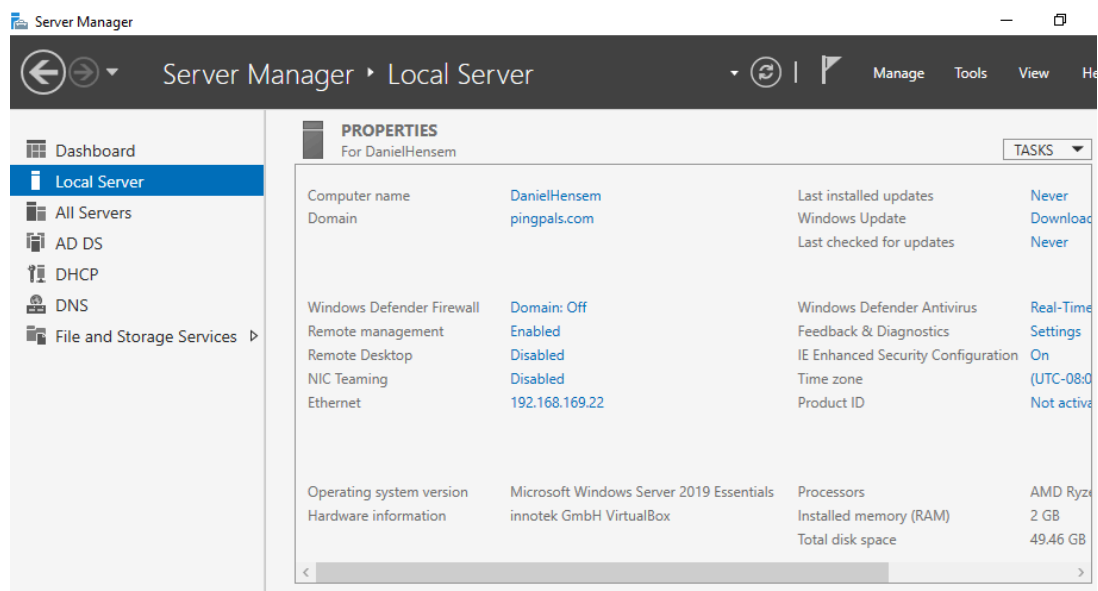
- 3.2.5. Show an image of a GNS3 VM running in VirtualBox. GNS3 server version (2.2.54) running on Ubuntu 20.04 within a VirtualBox environment, assigned IP address 192.168.154.3 and access credentials for both SSH and Web-UI.



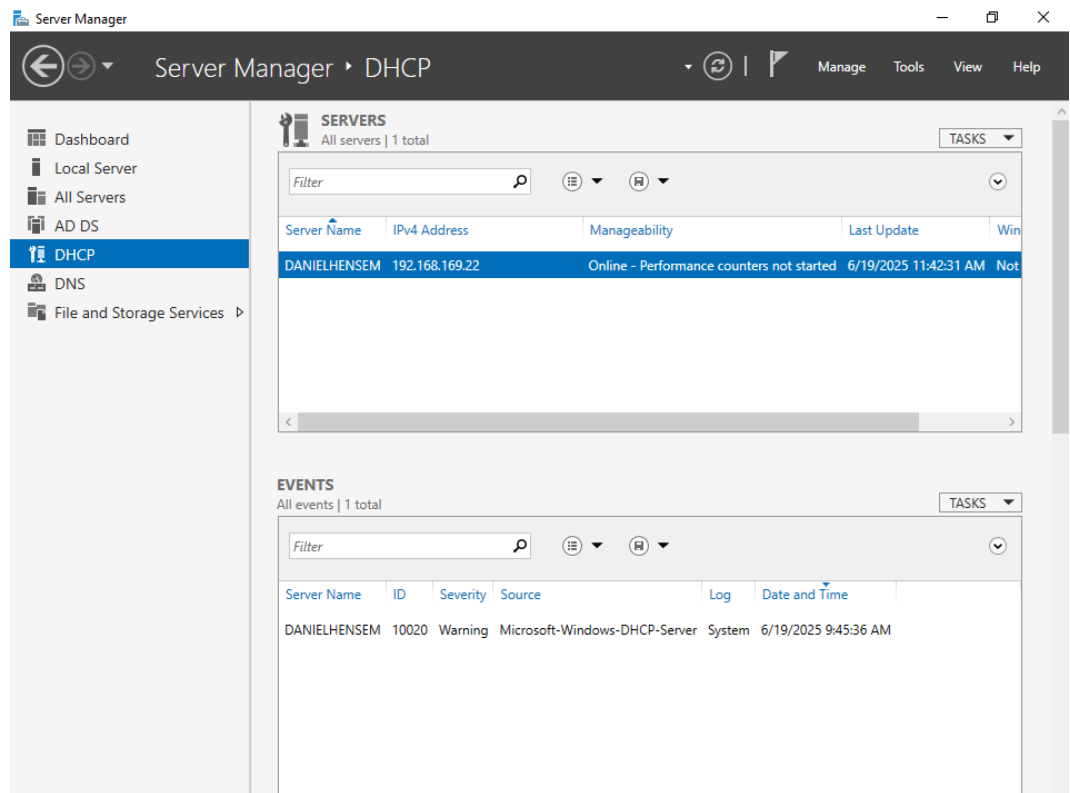
- 3.2.6. Show an image of the Windows Server 2019 Dashboard displaying installed roles and services, including Active Directory Domain Services (AD DS), DHCP, DNS, and File and Storage Services. These roles are essential for managing IP address allocation, domain-based authentication, name resolution, and shared storage, supporting the overall functionality of the integrated VirtualBox and GNS3 network infrastructure.



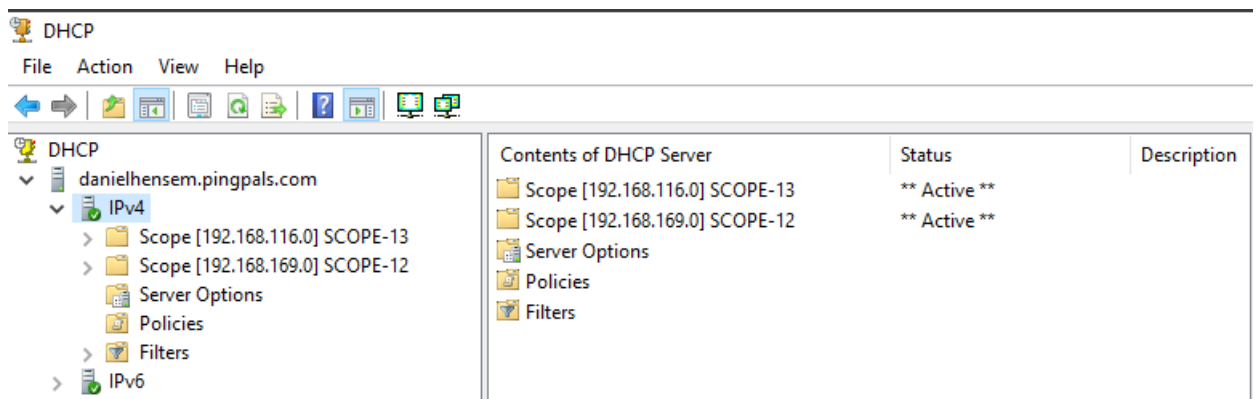
- 3.2.7. Show an image of the “Add Roles and Features Wizard” in Windows Server, displaying the selection of server roles available for installation such as Active Directory Domain Services, DHCP Server, and DNS Server have been installed, supporting network authentication, IP address management, and name resolution within the simulated environment.



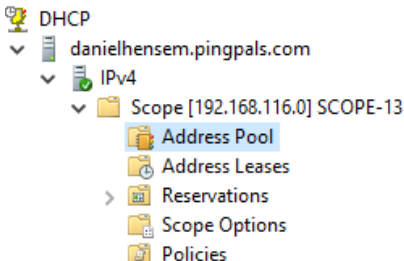
- 3.2.8. Show an image of Local Server properties in Windows Server Manager, it displays key system configurations such as computer name, domain membership, network settings, and system status. Domain pingpals.com and is assigned the IP address 192.168.169.22.



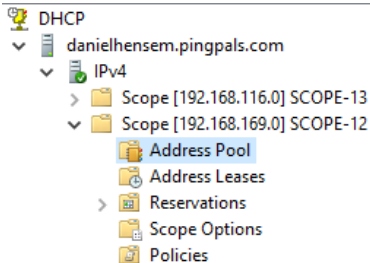
- 3.2.9. Show an image of the DHCP Server Manager interface in Windows Server which displays the DHCP server status, assigned IP address, and recent system events. DHCP service on the server DANIELHENSEM with IP address 192.168.169.22 is currently online and manageable.



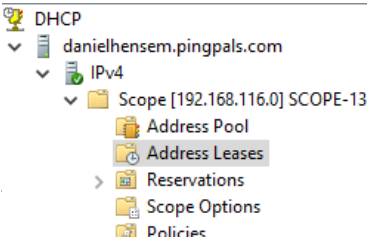
3.2.10. Show an image of the DHCP management console displaying active IPv4 scopes, it shows two active scopes which are 192.168.116.0 and 192.168.169.0, representing separate network segments for different departments within the simulated environment. Each scope is configured to automatically assign IP addresses to clients within its respective subnet.

	Start IP Address	End IP Address	Description
	192.168.116.100	192.168.116.254	Address range for distribution

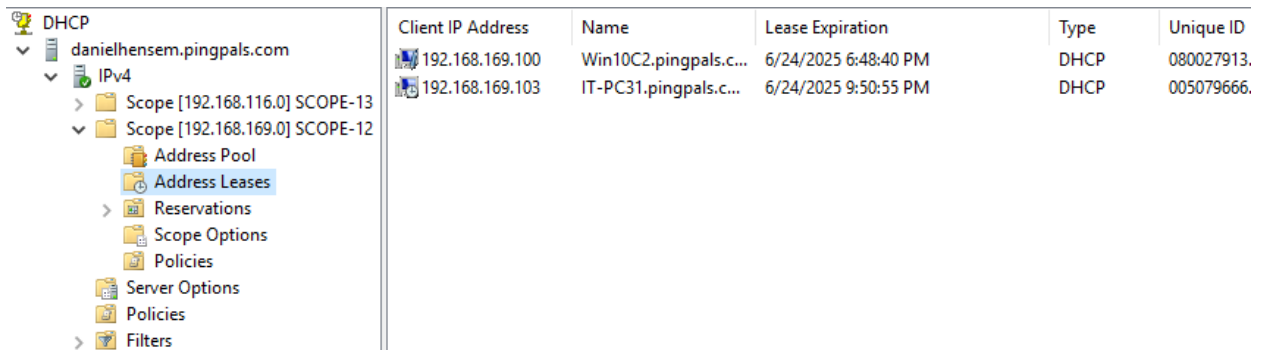
3.2.11. Show an image of the DHCP address pool configuration for the 192.168.116.0 scope, the address pool starts at 192.168.116.100 and ends at 192.168.116.254, specifying the range of client IPs that can be leased within this subnet.

	Start IP Address	End IP Address	Description
	192.168.169.100	192.168.169.254	Address range for distribution

3.2.12. Show an image of the DHCP address pool configuration for the 192.168.169.0 scope, the address pool starts at 192.168.169.100 and ends at 192.168.169.254, specifying the range of client IPs that can be leased within this subnet.

	Client IP Address	Name	Lease Expiration	Type	Unique ID
	192.168.116.100	Admin-PC11.pingp...	6/19/2025 2:26:29 PM	DHCP	005079666
	192.168.116.101	WIN10C1.pingpals...	6/19/2025 2:27:22 PM	DHCP	080027315
	192.168.116.102	Admin-PC41.pingp...	6/19/2025 2:33:34 PM	DHCP	005079666
	192.168.116.103	Admin-PC21.pingp...	6/19/2025 2:34:00 PM	DHCP	005079666

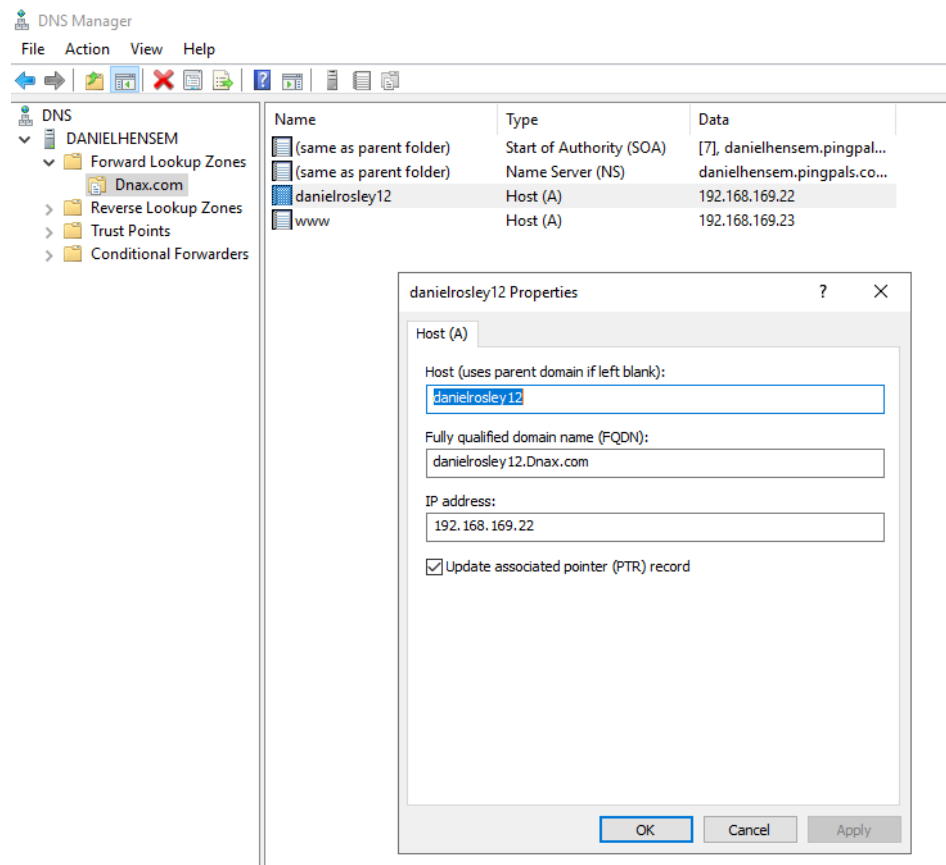
- 3.2.13. Show an image of the DHCP address lease list under the 192.168.116.0 scope, which provides a detailed view of IP addresses dynamically assigned to client devices within the network. Clients such as WIN10C1 and Admin-PC11 are shown to have received valid IP addresses through DHCP, confirming successful operation of the server in distributing and managing network configurations.



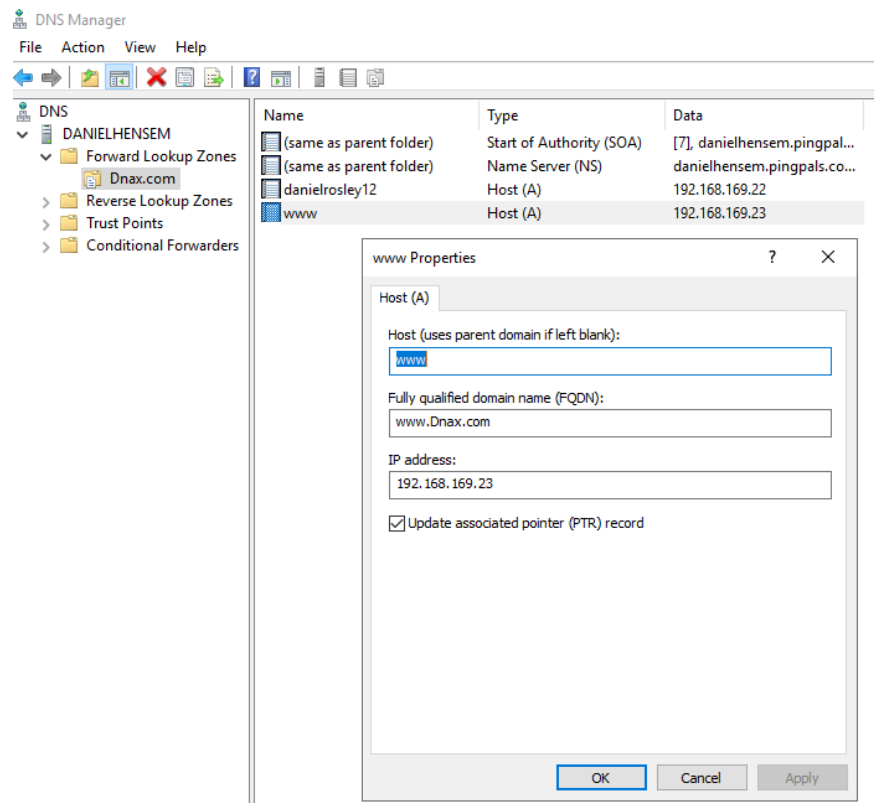
The screenshot shows the DHCP console interface. On the left, the tree view is expanded to 'Scope [192.168.116.0] SCOPE-13', with 'Address Leases' selected. The main pane displays a table of active leases.

Client IP Address	Name	Lease Expiration	Type	Unique ID
192.168.169.100	Win10C2.pingpals.c...	6/24/2025 6:48:40 PM	DHCP	080027913.
192.168.169.103	IT-PC31.pingpals.c...	6/24/2025 9:50:55 PM	DHCP	005079666.

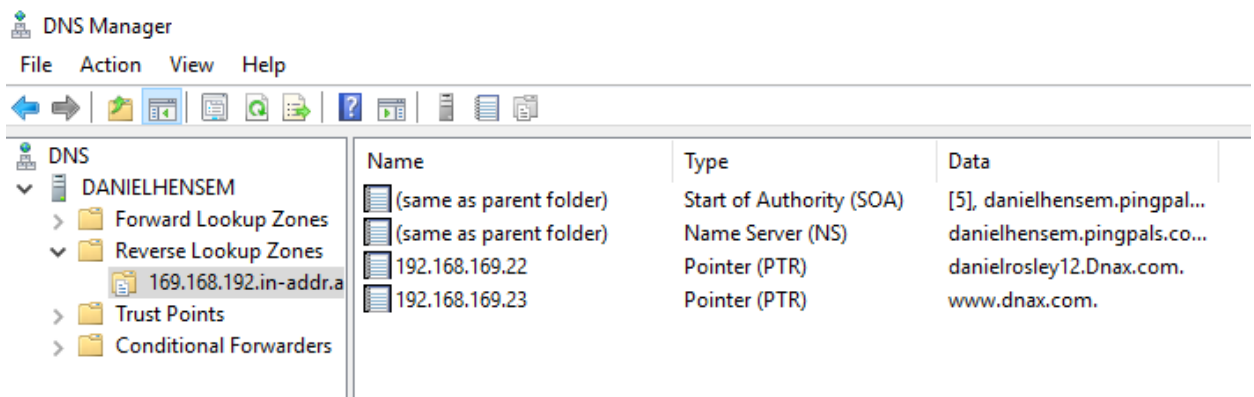
- 3.2.14. Show an image of the DHCP address lease list under the 192.168.169.0 scope, which provides a detailed view of IP addresses dynamically assigned to client devices within the network. Clients such as WIN10C2 and IT-PC31 are shown to have received valid IP addresses through DHCP, confirming successful operation of the server in distributing and managing network configurations.



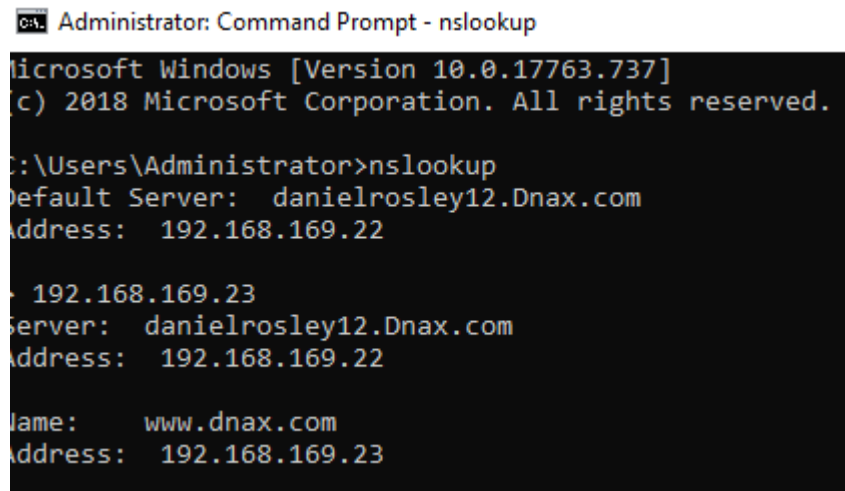
- 3.2.15. Show an image of the DNS Manager interface displaying a configured Host (A) record, it shows danielrosley12 host entry, mapped to the IP address 192.168.169.22, with a fully qualified domain name (FQDN) of danielrosley12.Dnax.com.



- 3.2.16. Show an image of the DNS Manager interface with a configured Host (A) record for a web server, the `www` entry under the `Dnax.com` forward lookup zone, linked to the IP address `192.168.169.23`, forming the fully qualified domain name (FQDN) `www.Dnax.com`.



- 3.2.17. Show an image of the DNS Manager displaying reverse lookup zone records, which demonstrates the use of Pointer (PTR) records for IP-to-hostname resolution. It shows the 169.168.192.in-addr.arpa reverse lookup zone, containing PTR records for IP addresses 192.168.169.22 and 192.168.169.23, corresponding to danielrosley12.Dnax.com and www.dnax.com, respectively.



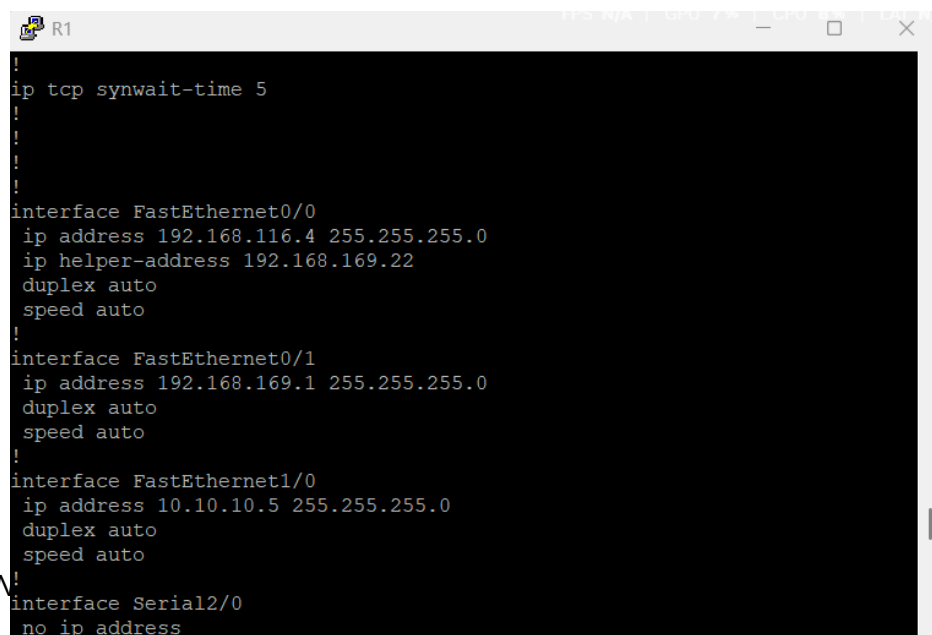
```
Administrator: Command Prompt - nslookup
Microsoft Windows [Version 10.0.17763.737]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>nslookup
Default Server:  danielrosley12.Dnax.com
Address:  192.168.169.22

192.168.169.23
Server:  danielrosley12.Dnax.com
Address:  192.168.169.22

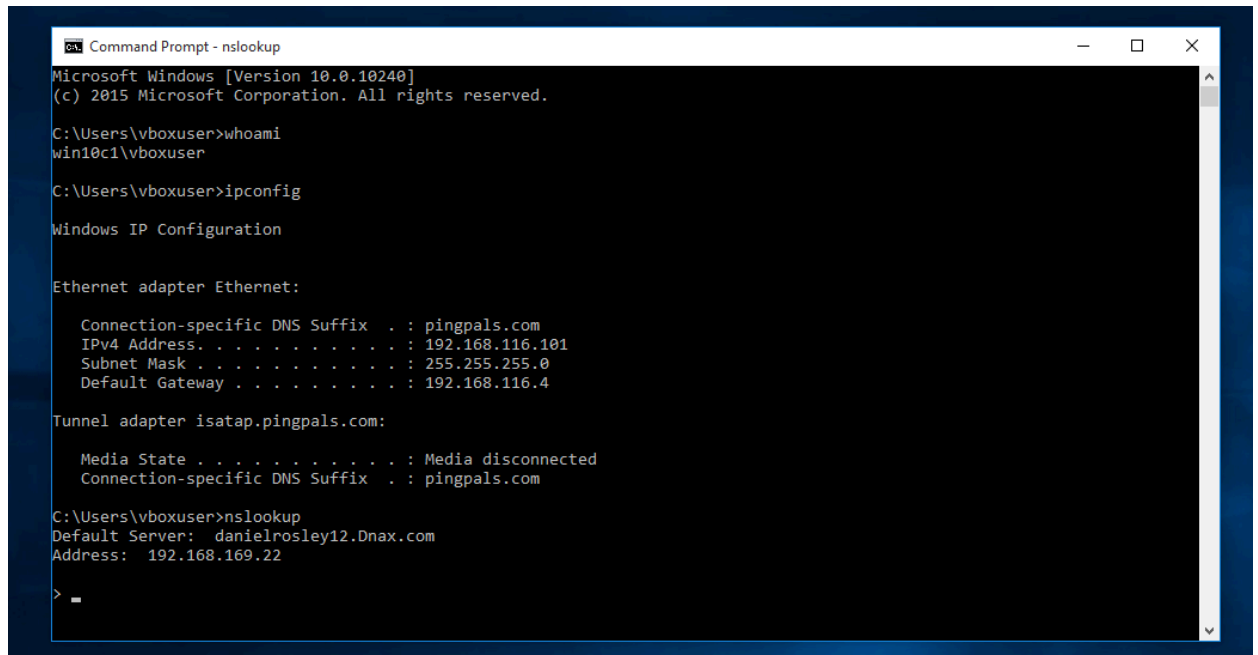
Name:    www.dnax.com
Address:  192.168.169.23
```

- 3.2.18. Show an image of the Command Prompt on Windows Server performing an nslookup test, which verifies the functionality of the configured DNS server. It displays a successful name resolution where the DNS server danielrosley12.dnax.com (IP: 192.168.169.22) responds to a query for www.dnax.com and returns the correct IP address 192.168.169.23.



```
R1
ipconfig
ip tcp synwait-time 5
!
!
!
!
!
interface FastEthernet0/0
 ip address 192.168.116.4 255.255.255.0
 ip helper-address 192.168.169.22
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 192.168.169.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 10.10.10.5 255.255.255.0
 duplex auto
 speed auto
!
interface Serial2/0
 no ip address
```

- 3.2.19. Show an image of the router interface configuration(R1), configuration of three FastEthernet interfaces which is FastEthernet0/0 with IP address 192.168.116.4, FastEthernet0/1 with IP address 192.168.169.1, and FastEthernet1/0 with IP 10.10.10.5. Additionally, the command ip helper-address 192.168.169.22 on interface FastEthernet0/0 forwards DHCP broadcast requests to the DHCP server located in another subnet.



```
Command Prompt - nslookup
Microsoft Windows [Version 10.0.10240]
(c) 2015 Microsoft Corporation. All rights reserved.

C:\Users\vboxuser>whoami
win10c1\vboxuser

C:\Users\vboxuser>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : pingpals.com
    IPv4 Address. . . . . : 192.168.116.101
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.116.4

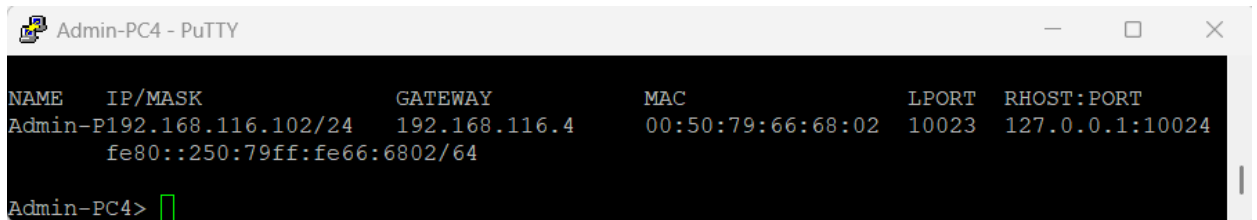
Tunnel adapter isatap.pingpals.com:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : pingpals.com

C:\Users\vboxuser>nslookup
Default Server:  danielrosley12.Dnax.com
Address:  192.168.169.22

>
```

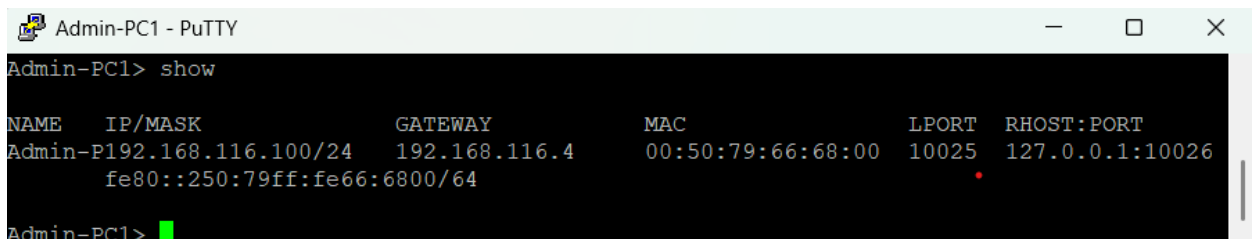
- 3.2.20. Show an image of the Command Prompt from Agent1 displaying network configuration and DNS resolution. Result of the ipconfig command, indicating that Agent1 (host WIN10C1) received an IP address of 192.168.116.101, a subnet mask of 255.255.255.0, and a default gateway of 192.168.116.4. The subsequent nslookup confirms that the default DNS server is danielrosley12.dnax.com with IP address 192.168.169.22.



```
Admin-PC4 - PuTTY
NAME      IP/MASK      GATEWAY      MAC      LPORT      RHOST:PORT
Admin-P1  192.168.116.102/24  192.168.116.4  00:50:79:66:68:02  10023  127.0.0.1:10024
          fe80::250:79ff:fe66:6802/64
Admin-PC4> 
```

3.2.21. Show an image of the network configuration on Admin-PC4 accessed via PuTTY.

Admin-PC4 is assigned the IP address 192.168.116.102/24 with a default gateway of 192.168.116.4. The MAC address is listed as 00:50:79:66:68:02, and the client is connected using local port 10023.



```
Admin-PC1 - PuTTY
Admin-PC1> show
NAME      IP/MASK      GATEWAY      MAC      LPORT      RHOST:PORT
Admin-P1  192.168.116.100/24  192.168.116.4  00:50:79:66:68:00  10025  127.0.0.1:10026
          fe80::250:79ff:fe66:6800/64
Admin-PC1> 
```

3.2.22. Show an image of the network configuration on Admin-PC1 accessed via PuTTY.

Admin-PC1 has received the IP address 192.168.116.100/24, with a default gateway of 192.168.116.4. The MAC address is 00:50:79:66:68:00, and the device is connected through local port 10025.

```
Admin-PC2 - PuTTY

NAME      IP/MASK      GATEWAY      MAC      LPORT      RHOST:PORT
Admin-P1  192.168.116.103/24  192.168.116.4  00:50:79:66:68:03  10027  127.0.0.1:10028
          fe80::250:79ff:fe66:6803/64

Admin-PC2> █
```

3.2.23. Show an image of the network configuration on Admin-PC2 accessed via PuTTY.

Admin-PC2 is assigned the IP address 192.168.116.103/24 with a default gateway of 192.168.116.4. The MAC address is 00:50:79:66:68:03, and the local port used is 10027.

```
IT-PC3 - PuTTY

PC1> show

NAME      IP/MASK      GATEWAY      MAC      LPORT      RHOST:PORT
PC1       192.168.169.104/24  192.168.169.1  00:50:79:66:68:04  10033  127.0.0.1:10034
          fe80::250:79ff:fe66:6804/64
```

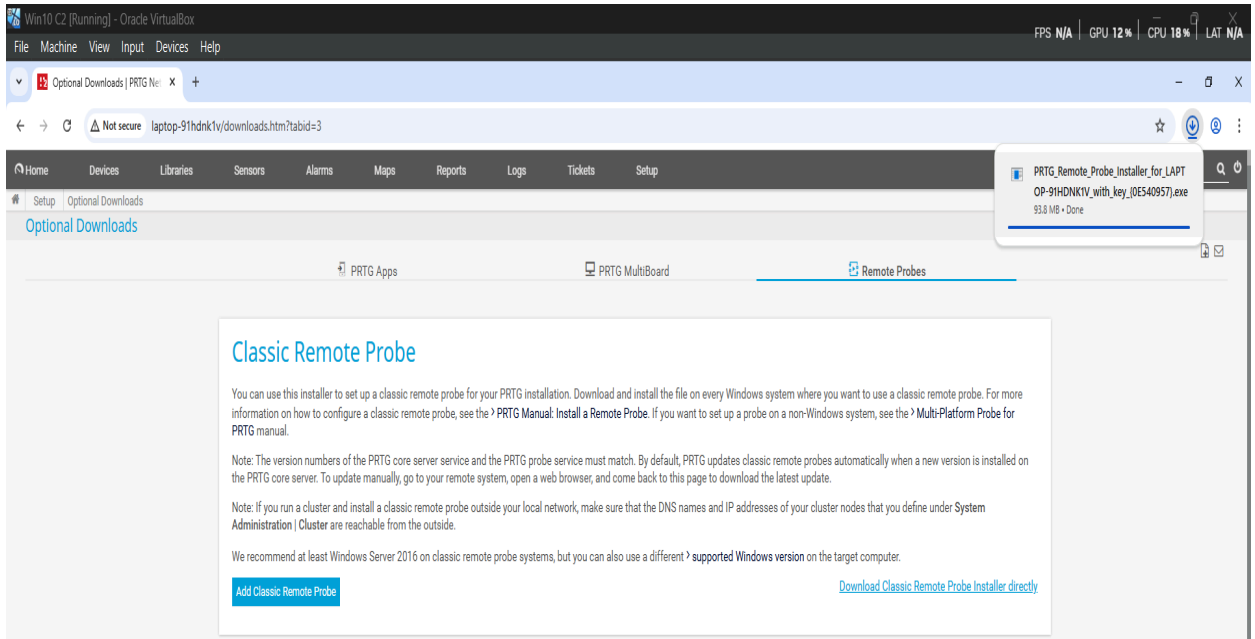
3.2.24. Show an image of the network configuration on IT-PC3 accessed via PuTTY.

IT-PC3, labeled as PC1, is assigned the IP address 192.168.169.104/24 with the default gateway 192.168.169.1. The MAC address is 00:50:79:66:68:04, and the connection is established through local port 10033.

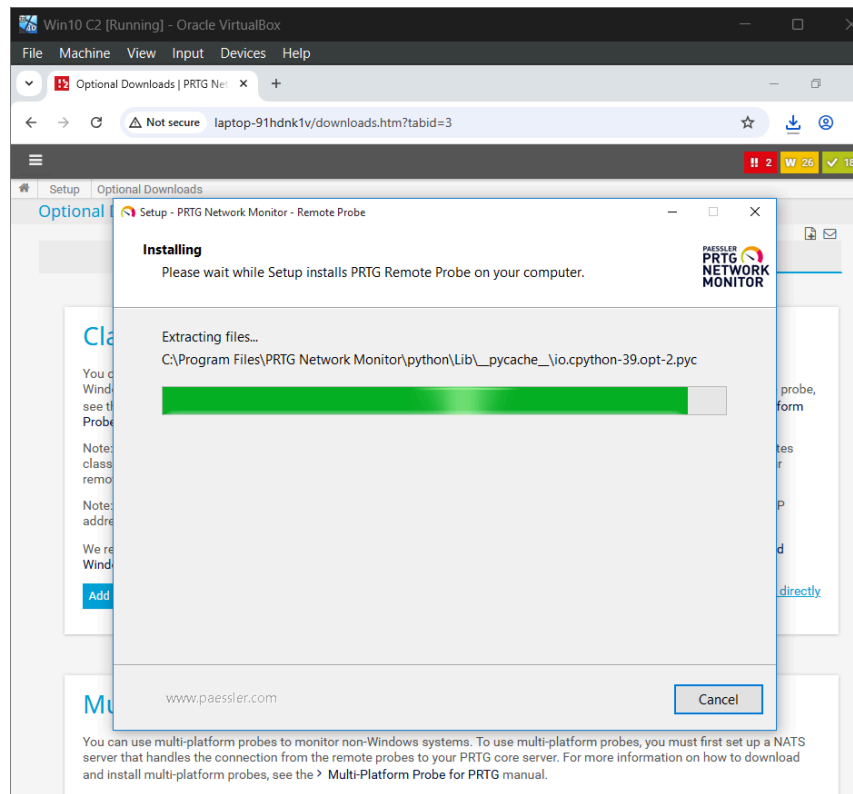
3.3. Setup and Configuration of PRTG



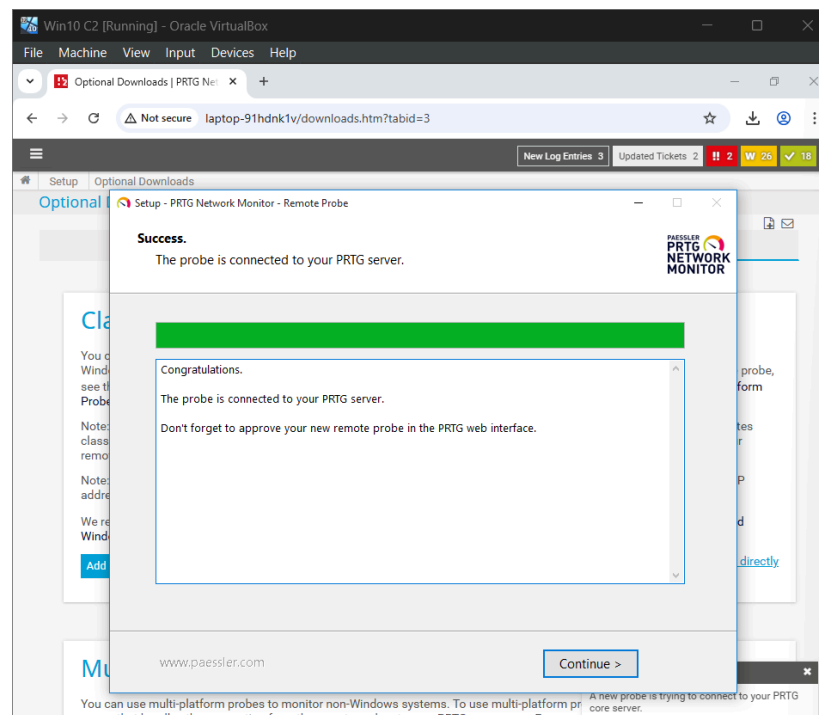
- 3.3.1. Show an image of the PRTG Network Monitor login page accessed from localhost. PRTG is hosted locally on the machine identified as LAPTOP-91HDNK1V and is accessible via the loopback address 127.0.0.1.



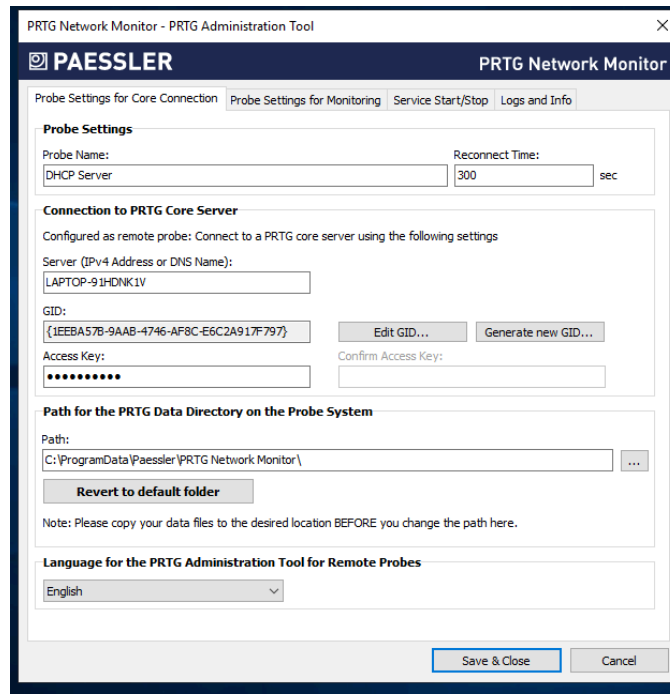
- 3.3.2. Show an image of the PRTG Network Monitor interface accessed from a client PC for downloading the monitoring agent. The Classic Remote Probe is downloaded to enable the client to participate in network monitoring hosted on the localhost.



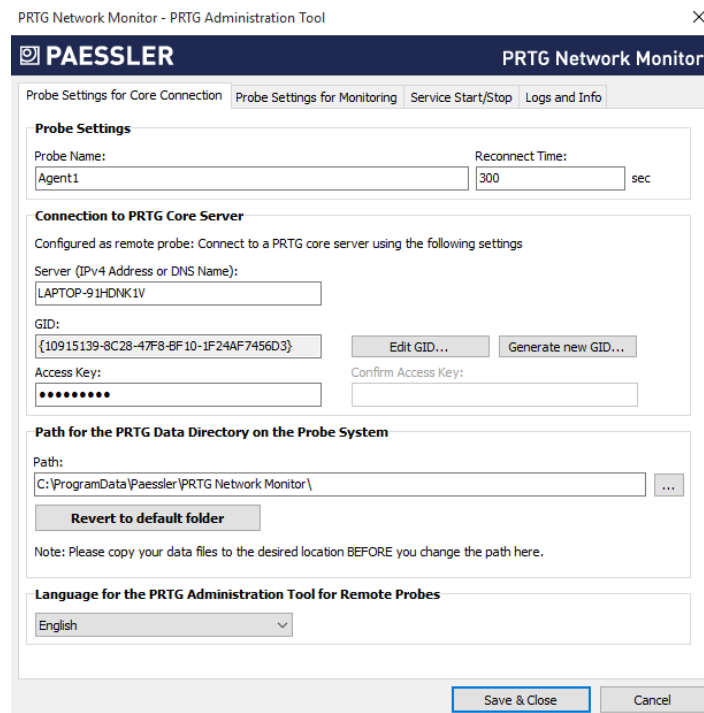
3.3.3. Show an image of the PRTG Network Monitor being installed in client PCs for monitoring.



3.3.4. Show an image of the PRTG Network Monitor successfully installed and connected to the PRTG server(LocalHost).



- 3.3.5. Show an image of the PRTG Administration Tool displaying probe configuration settings, which show the probe is named "DHCP Server" and is associated with the local host LAPTOP-91HDNK1V.

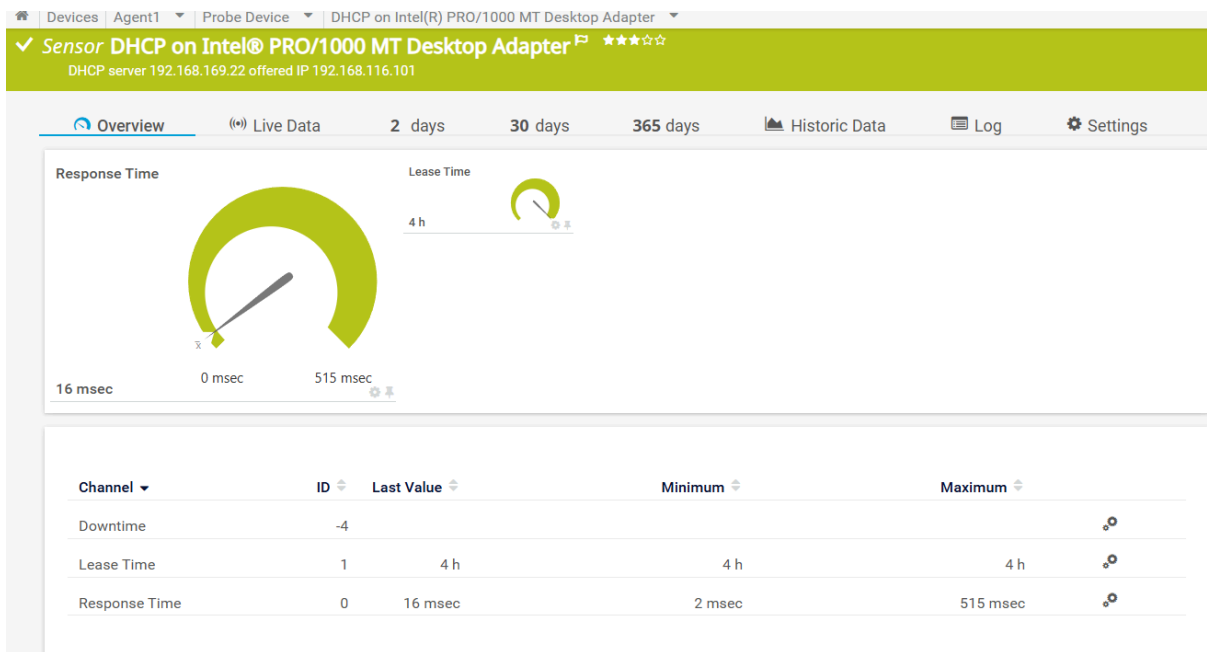


- 3.3.6. Show an image of the PRTG Administration Tool displaying probe configuration settings, which show the probe is named "Agent1" and is associated with the local host LAPTOP-91HDNK1V.

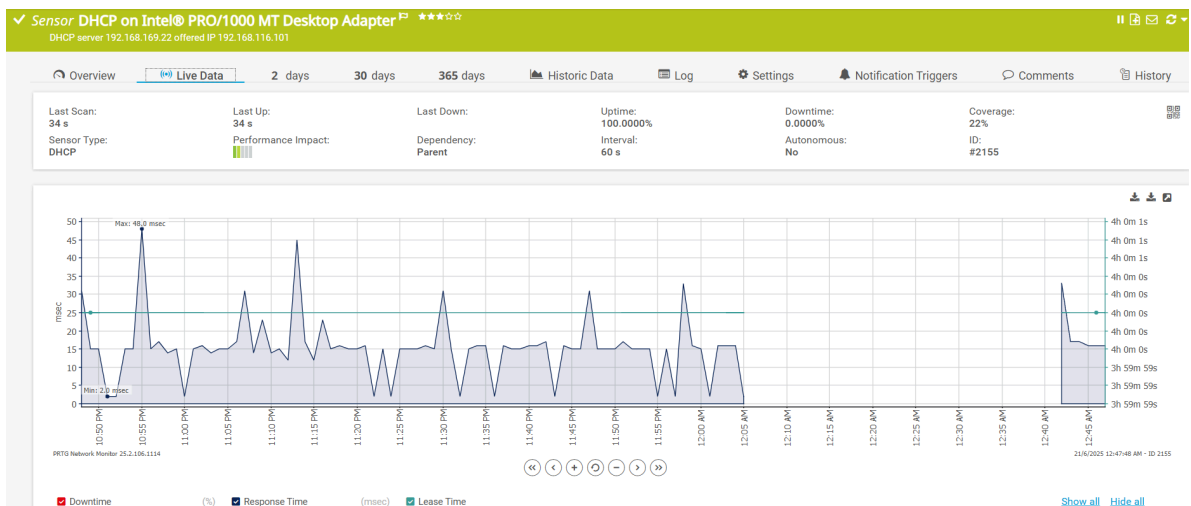
4. Evaluation and Results

4.1. PRTG Monitoring Scenarios

4.1.1. DHCP Service Monitoring



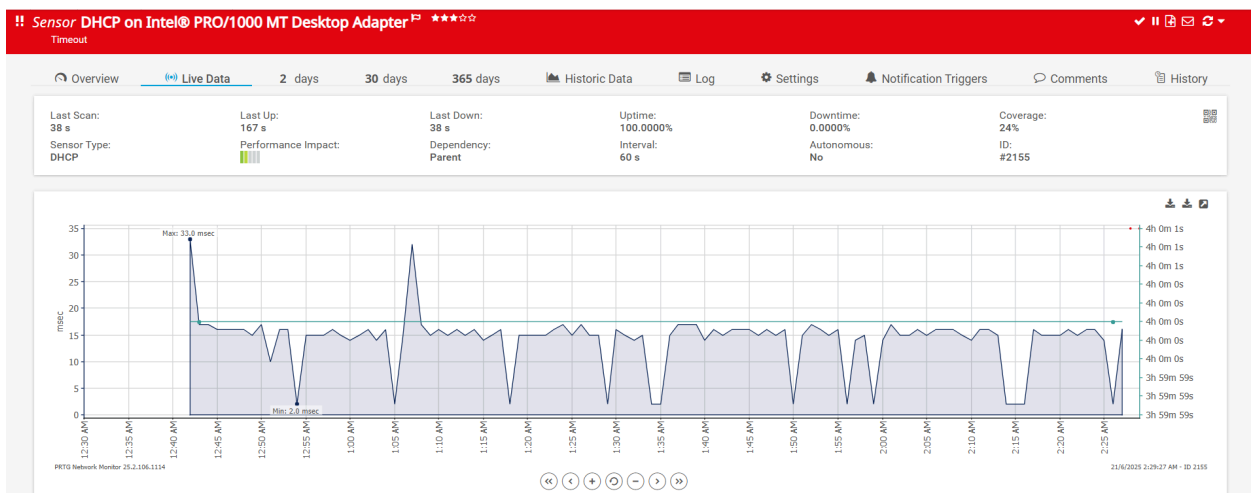
4.1.1.1. Show an image of the DHCP service on Agent1 with response and lease times from server 192.168.169.22.



4.1.1.2. Show an image of the DHCP sensor's live data graph displaying response and lease time fluctuations from server 192.168.169.22.

SERVICES					TASKS ▼
All services 1 total					
Filter					
Server Name	Display Name	Service Name	Status	Start Type	
DANIELHENSEM	DHCP Server	DHCPServer	Paused	Automatic	

4.1.1.3. Show an image of the DHCP service status on the server, indicating that the DHCPServer service is currently paused and set to start automatically.

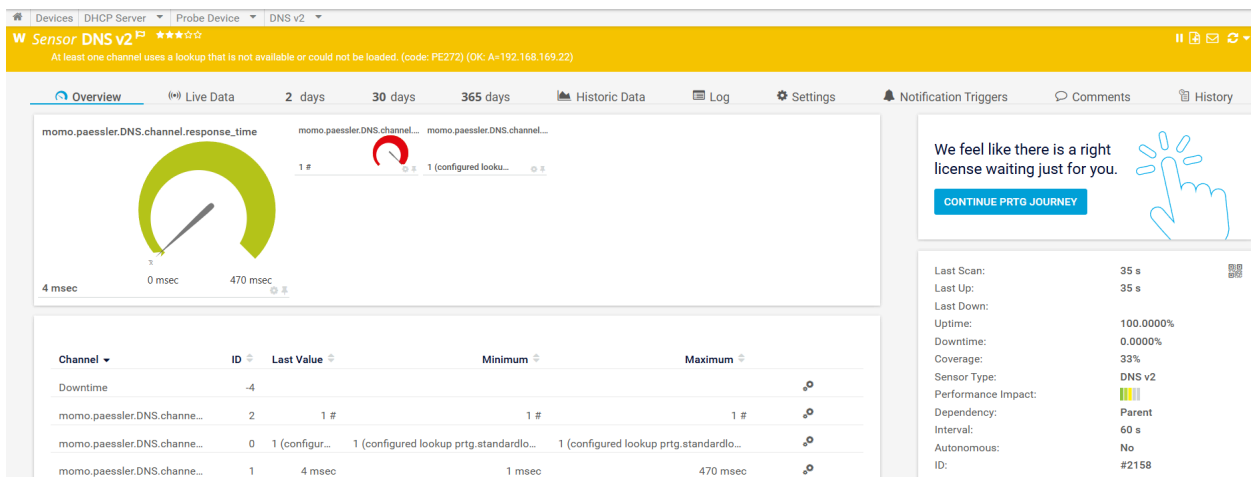


4.1.1.4. Show an image of the DHCP sensor in a timeout state on PRTG, indicating interrupted DHCP communication or service failure, as shown by the red alert and missing response times.

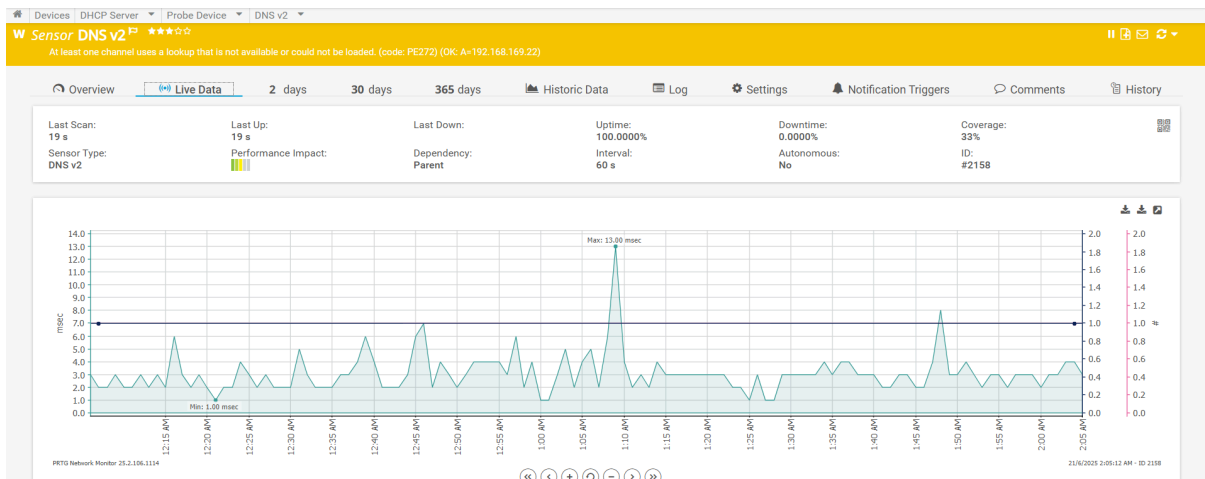
Date Time ^	Response Time ^	Lease Time ^	Downtime ^	Coverage ^
21/6/2025 2:30:49 AM	Error	Error	Error	100 %
21/6/2025 2:29:49 AM	Error	Error	Error	100 %
21/6/2025 2:28:49 AM	Error	Error	Error	100 %
21/6/2025 2:28:00 AM	Error	Error	Error	100 %
21/6/2025 2:27:49 AM	Error	Error	Error	100 %

- 4.1.1.5. Show an image of the DHCP sensor error log in PRTG, it displays repeated failure entries for response time, lease time, and downtime indicating a persistent issue with DHCP service delivery.

4.1.2. DNS Service Monitoring



- 4.1.2.1. Show an image of the DNS sensor overview showing response time and query status in PRTG.



4.1.2.2. Show an image of the DNS sensor live data graph indicating query performance over time.

SERVICES
All services | 1 total

Filter

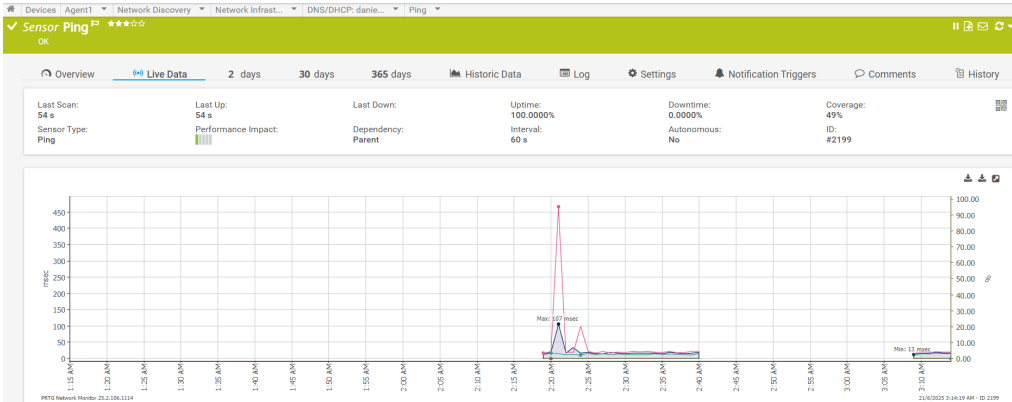
Server Name	Display Name	Service Name	Status	Start Type
DANIELHENSEM	DNS Server	DNS	Paused	Automatic

4.1.2.3. Show an image of the DNS service status from the server, displaying it as Paused with Automatic startup type.

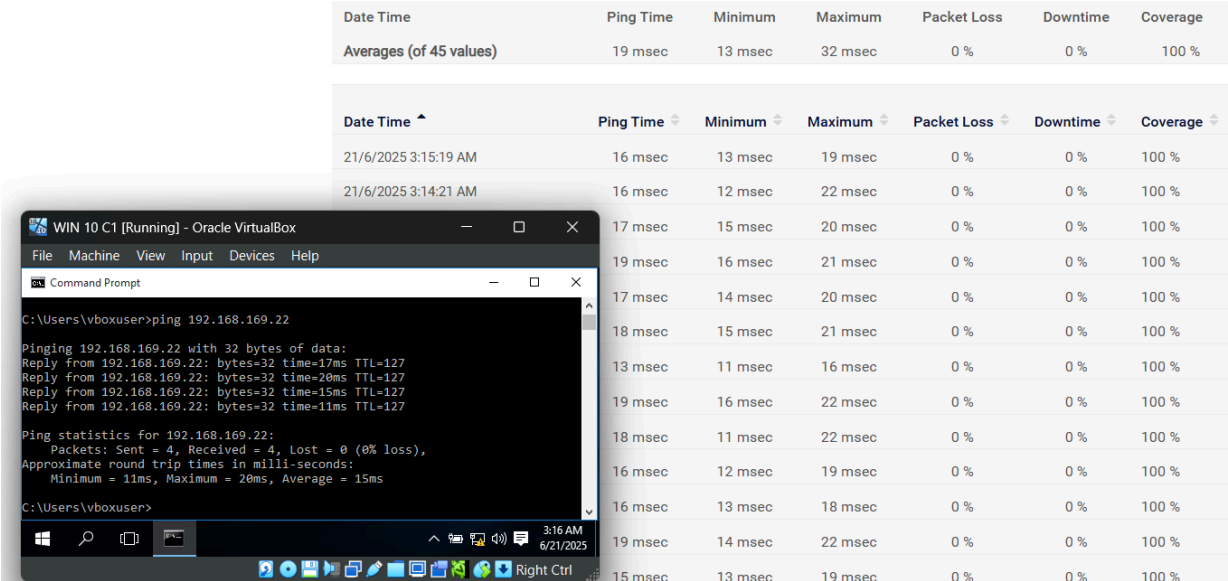
Date Time	momo.paessler.DNS.channel.records_resolved	momo.paessler.DNS.channel.response_time	momo.paessler.DNS.channel.record_count	Downtime
Averages (of 129 values)		24 msec	1 #	2 %
Date Time ^	momo.paessler.DNS.channel.records_resolved	momo.paessler.DNS.channel.response_time	momo.paessler.DNS.channel.record_count	Downtime
21/6/2025 2:40:35 AM	Error	Error	Error	Error
21/6/2025 2:40:01 AM	Error	Error	Error	Error
21/6/2025 2:39:05 AM	1 (configured lookup prtg.standardlookups.paessler.dns.lookup_records_found is empty or not available)	17 msec	1 #	0 %
21/6/2025 2:38:05 AM	1 (configured lookup prtg.standardlookups.paessler.dns.lookup_records_found is empty or not available)	21 msec	1 #	0 %
21/6/2025 2:37:05 AM	1 (configured lookup prtg.standardlookups.paessler.dns.lookup_records_found is empty or not available)	15 msec	1 #	0 %

4.1.2.4. Show an image of the DNS monitoring table showing errors in record resolution, response time, and downtime logs.

4.1.3. ICMP Service Monitoring

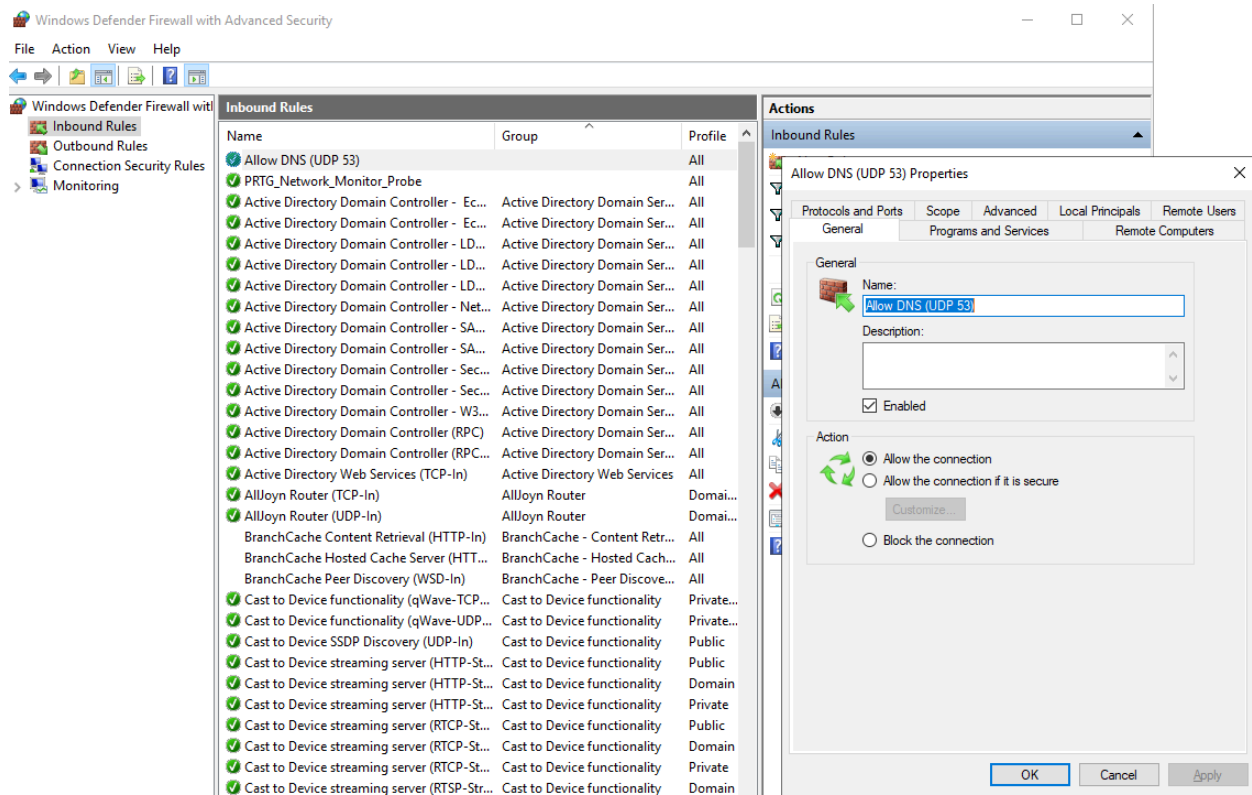


4.1.3.1. Show an image of the ICMP (Ping) monitoring graph, displaying live latency performance trends over time

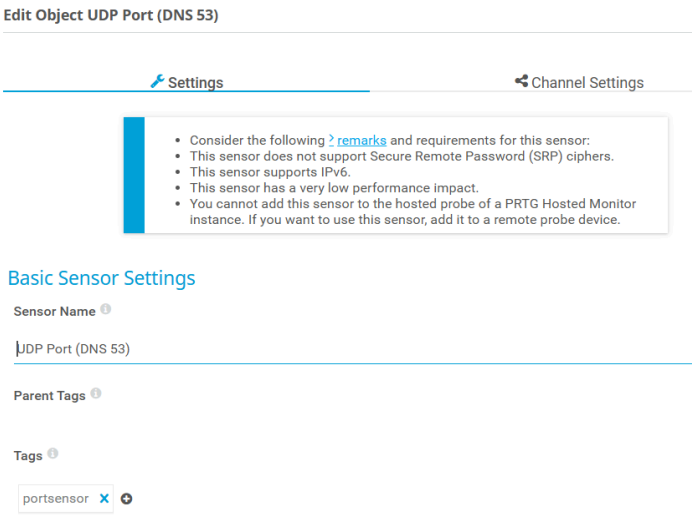


4.1.3.2. Show an image of the ICMP ping result from a client device and its detailed statistics recorded in PRTG, including ping time, packet loss, and response metrics.

4.1.4. UDP Service Monitoring

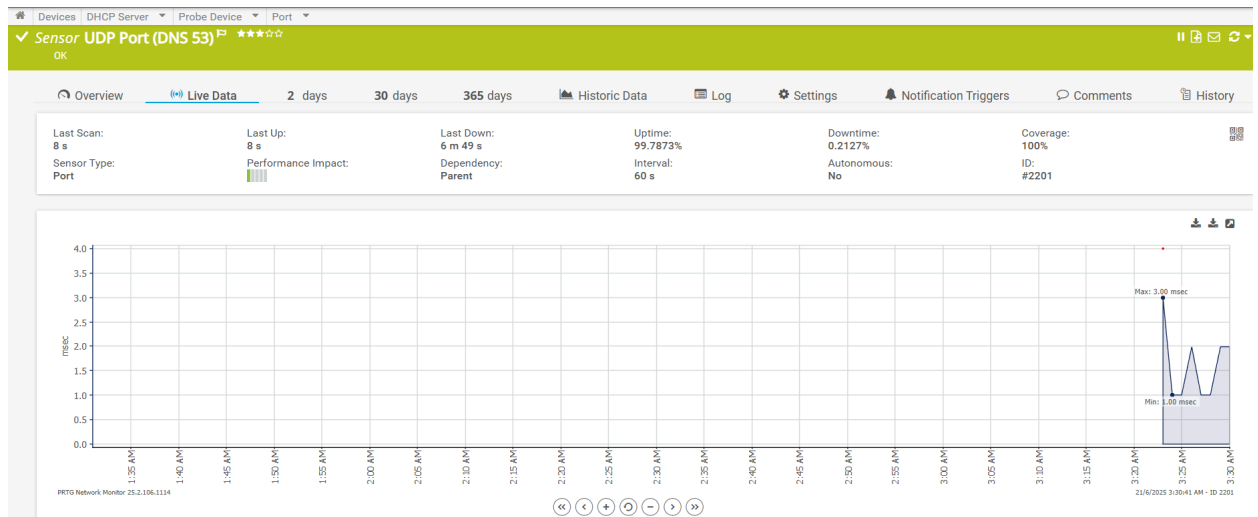


4.1.4.1. Show an image of the Windows Defender Firewall configuration, where the UDP port (53) is allowed for DNS monitoring—enabling UDP traffic detection by



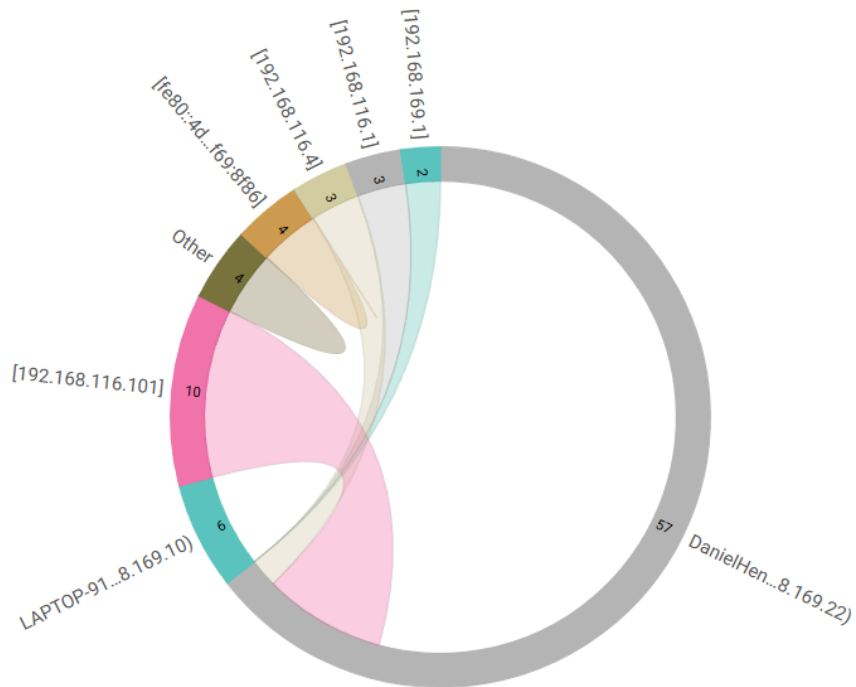
PRTG.

4.1.4.2. Show an image of the UDP Port sensor settings in PRTG, where DNS port 53 is monitored to track incoming UDP traffic activity.



4.1.4.3. Show an image of the live data graph for the UDP Port (DNS 53) sensor in PRTG, displaying real-time packet activity over the monitoring period.

4.1.5. Packet Sniffer Service Monitoring



- 4.1.5.1. Show an image of the packet sniffer traffic distribution chart, illustrating the data communication between monitored devices, including source and destination IPs and the amount of traffic exchanged.

Pos	Source IP	Destination IP	Bytes	
1.	DanielHensem.pingpals.com (192.168.169.22)	LAPTOP-91HDNK1V (192.168.169.10)	74 KB	48 %
2.	[192.168.116.101]	DanielHensem.pingpals.com (192.168.169.22)	16 KB	10 %
3.	DanielHensem.pingpals.com (192.168.169.22)	[192.168.116.101]	11 KB	7 %
4.	LAPTOP-91HDNK1V (192.168.169.10)	DanielHensem.pingpals.com (192.168.169.22)	8,980 Byte	6 %
Other			6,167 Byte	4 %
5.	[fe80::4d44:636a:cf69:8f86]	[ff02::fb]	5,760 Byte	4 %
6.	[192.168.116.4]	DanielHensem.pingpals.com (192.168.169.22)	4,720 Byte	3 %
7.	[192.168.116.1]	DanielHensem.pingpals.com (192.168.169.22)	4,659 Byte	3 %
8.	LAPTOP-91HDNK1V (192.168.169.10)	[192.168.169.255]	4,659 Byte	3 %
9.	LAPTOP-91HDNK1V (192.168.169.10)	[224.0.0.251]	4,480 Byte	3 %
10.	[192.168.169.1]	DanielHensem.pingpals.com (192.168.169.22)	3,290 Byte	2 %
11.	DanielHensem.pingpals.com (192.168.169.22)	[192.168.116.4]	2,736 Byte	2 %
12.	[fe80::4d44:636a:cf69:8f86]	[ff02::1:3]	2,688 Byte	2 %
13.	DanielHensem.pingpals.com (192.168.169.22)	[192.203.230.10]	2,435 Byte	2 %
14.	LAPTOP-91HDNK1V (192.168.169.10)	[224.0.0.252]	2,048 Byte	1 %
15.	DanielHensem.pingpals.com (192.168.169.22)	[192.112.36.4]	1,373 Byte	< 1 %
16.	DanielHensem.pingpals.com (192.168.169.22)	[192.168.169.255]	243 Byte	< 1 %

4.1.5.2. Show an image of the detailed packet sniffer table showing traffic sources, destinations, and the volume of data (in bytes), identifying which devices are generating the most network activity.

4.2. Analysis of Monitoring Data

Monitoring process utilized PRTG Network Monitor to evaluate critical network services and overall traffic behavior within the SME infrastructure. This analysis is based on the collected metrics and visual outputs from several configured sensors, including DHCP, DNS, ICMP (Ping), UDP, and Packet Sniffer.

4.2.1. DHCP Service Monitoring

DHCP sensor provided data on response time, lease time, and service uptime. During the monitoring period, it was observed that the DHCP server occasionally returned errors, particularly when the service was paused, as indicated in Figure 4.1.1.3. While lease times were generally stable at approximately 4 hours, moments of service failure resulted in an absence of IP lease allocation, disrupting client device connectivity. These errors were evident in the tabular view (Figure 4.1.1.5), where all response metrics were listed as "Error," signaling a complete breakdown in service response during that time frame.

4.2.2. DNS Service Monitoring

DNS sensor focused on response time, records resolved, and the frequency of DNS lookup failures. DNS performance was generally acceptable, with an average response time of 24 milliseconds. However, there were several periods where DNS queries failed, indicated by red error messages and failure to resolve configured lookups (Figure 4.1.2.4). These failures may point to issues with the local DNS service being paused or temporary unavailability of the lookup targets. Despite occasional errors, DNS uptime remained above 99%, suggesting that issues were intermittent rather than persistent.

4.2.3. ICMP (Ping) Monitoring

Ping sensor was used to measure network availability and response latency between hosts. Results show an average ping time of 19 milliseconds with no packet loss throughout the monitoring period (Figure 4.1.3.1). These results indicate a reliable and responsive network. The command-line ping output (Figure 4.1.3.2) further validated these results by confirming consistent ICMP echo replies with stable round-trip times.

4.2.4. UDP Port Monitoring

To evaluate services operating over the UDP protocol, a UDP port sensor was configured to monitor DNS over port 53. This sensor recorded minimal delay and consistent activity (Figure 4.1.4.3), indicating that UDP-based DNS queries were being successfully transmitted and responded. The firewall configuration (Figure 4.1.4.1) showed that the relevant port (UDP 53) was allowed, ensuring that traffic was not being blocked at the host level.

4.2.5. Packet Sniffer Monitoring

Packet sniffing data provided a broader view of bandwidth usage and traffic flow within the network. Visual output (Figure 4.1.5.1) illustrated that the majority of traffic originated from DanielHensem.pingpals.com (192.168.169.22), followed by LAPTOP-91HDNK1V (192.168.169.10)

and 192.168.116.101. The corresponding data table (Figure 4.1.5.2) revealed that 48% of network traffic came from a single node, which may indicate high resource consumption or centralized traffic handling. This information is critical for traffic engineering and capacity planning.

4.3. Observations and Performance

- Network Uptime and Reachability

Networks show strong uptime and availability, particularly highlighted by the ICMP (Ping) sensor which showed no packet loss and consistent latency. It shows that core network connectivity is stable and reliable, a critical factor for uninterrupted business operations.

- DHCP and DNS Vulnerabilities

Both DHCP and DNS services showed a combination of stable performance and detectable issues during the monitoring process. For the DHCP service, total failures were observed when it was intentionally paused, resulting in IP address assignment problems and preventing new clients from connecting to the network. DNS monitoring revealed occasional resolution errors that may have affected access to websites and communication between devices. The services were intentionally paused to test whether PRTG could successfully detect and report these interruptions, which it did effectively.

- Service Configuration and Control

During the monitoring process, DHCP and DNS services were tested by pausing to evaluate the responsiveness and accuracy of the PRTG monitoring system. When DHCP service was halted, connected devices failed to obtain IP addresses, confirming that PRTG could detect and log service disruptions effectively. Pausing DNS service led to temporary name resolution failures, which were promptly reflected in the monitoring dashboard. These intentional interruptions helped validate PRTG's capability to identify service status changes in real time.

- Traffic Distribution Insights

Network usage was found to be concentrated among a few specific devices. The majority of data exchanges occurred between the DHCP/DNS server and selected clients, indicating a non-efficient share of bandwidth consumption. Certain nodes are consistently handling more traffic, which may affect overall performance if left unmanaged. Observation supports the need for implementing load balancing strategies, optimizing bandwidth allocation, and applying targeted policies to better manage traffic flow and maintain network efficiency.

- Protocol Monitoring Effectiveness

Use of sensors for both TCP/IP and UDP protocols, along with application-layer services like DHCP and DNS, proved to be effective in capturing a comprehensive view of the network. UDP Port 53 (DNS) traffic was clearly visible and measurable, demonstrating the sensor's ability to track stateless communication protocols.

5. Conclusion and Future Work

5.1. Summary of Findings

This study successfully demonstrated how PRTG Network Monitor can serve as an effective, affordable, and user-friendly monitoring solution for SMEs. By simulating a realistic small office network using GNS3, VirtualBox and PRTG, essential services like DHCP, DNS, ICMP and UDP were monitored in real time. The system accurately detected service downtimes, traffic anomalies and protocol-level performance issues. PRTG sensor-based design, SNMP integration and customizable alerts proved ideal for SMEs with limited IT expertise.

5.2. Recommendations for Future Improvements

Future work could explore:

1. Automation of alerts and response actions.
2. Enhanced security monitoring through IDS/firewall integration.
3. Scalability testing with more clients and services.
4. Mobile alerts for quicker incident response.
5. Training modules to empower non-technical SME staff.

This project highlights that with the right tools, SMEs can achieve enterprise-level network visibility without enterprise-level costs.

6. References

- Ahmed, M. I., & Noor, N. M. (2021). Design and implementation of a PRTG-based proactive network monitoring system for SMEs. *International Journal of Advanced Computer Science and Applications*, 12(6), 371–378. <https://doi.org/10.14569/IJACSA.2021.0120647>
- Alizan, A. N., & Zolkipli, M. F. (2024). A comparative study between Wireshark and Paessler Router Traffic Grapher (PRTG) in network monitoring and analysis. *INTI Journal*, 2024(1). <https://doi.org/10.61453/intij.202451>
- Fathima, A., & Devi, G. S. (2024). Enhancing university network management and security: A real-time monitoring, visualization & cyber attack detection approach using Paessler PRTG and Sophos Firewall. *International Journal of Systems Assurance Engineering and Management*. <https://doi.org/10.1007/s13198-024-02448-y>
- Li, X., & Zhang, L. (2022). A lightweight network monitoring framework for small enterprises using open-source tools. *Computer Networks*, 210, 109052. <https://doi.org/10.1016/j.comnet.2022.109052>
- Paessler AG. (2023). *PRTG Network Monitor: Monitoring for SMEs – Case studies and deployment best practices*. <https://www.paessler.com/solutions/sme>
- Sharma, P., & Bedi, R. (2023). Performance optimization of SME networks through SNMP-based monitoring and alerting systems. *Journal of Network and Computer Applications*, 222, 103470. <https://doi.org/10.1016/j.inca.2023.103470>
- Singh, R., & Kumari, S. (2020). Evaluating SNMP-based network monitoring solutions for educational and small business networks. *International Journal of Information Management Data Insights*, 1(1), 100004. <https://doi.org/10.1016/j.jjime.2020.100004>